

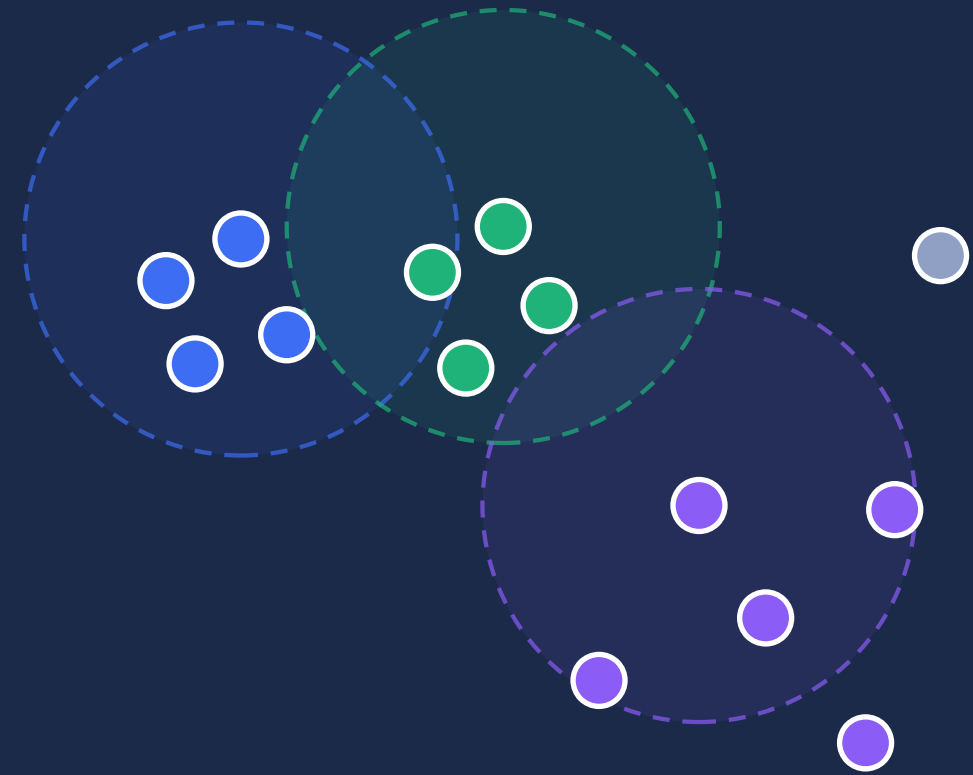
A STEP-BY-STEP LECTURE

OPTICS

Ordering Points To Identify
the Clustering Structure

Clustering groups with different densities

A lesson for high school students



Today's roadmap

1

Why OPTICS?

What goes wrong when we use only one radius.

2

Key ideas

Core distance and reachability distance.

3

The simulation

We run OPTICS step by step on 14 points.

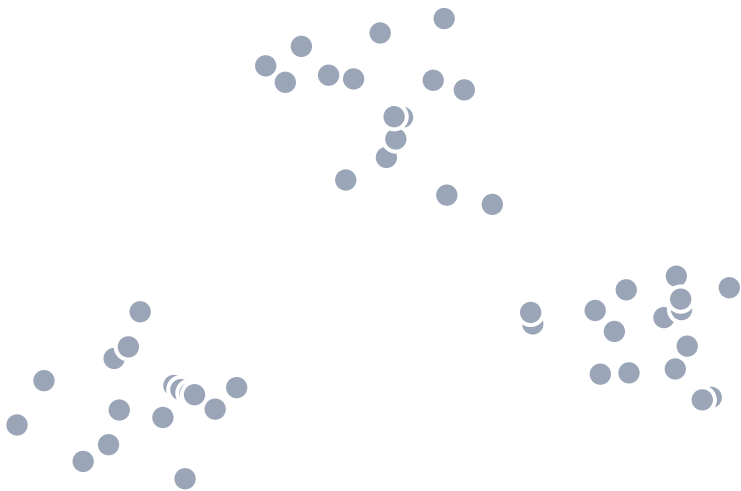
4

Reading the result

Turn a plot into clusters.

What is clustering?

Before: just dots



After: groups



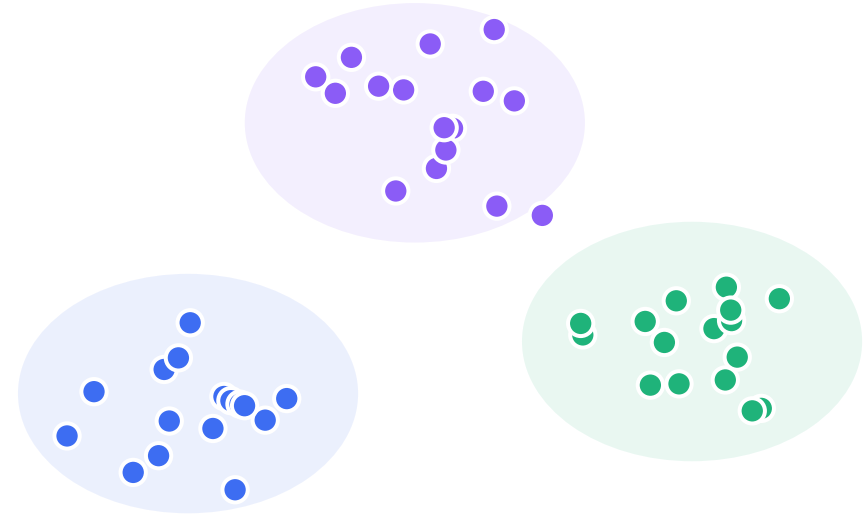
Clustering means sorting data points into groups of similar items. Nobody tells the computer the groups in advance. It has to find them by itself.

What is clustering?

Before: just dots



After: groups



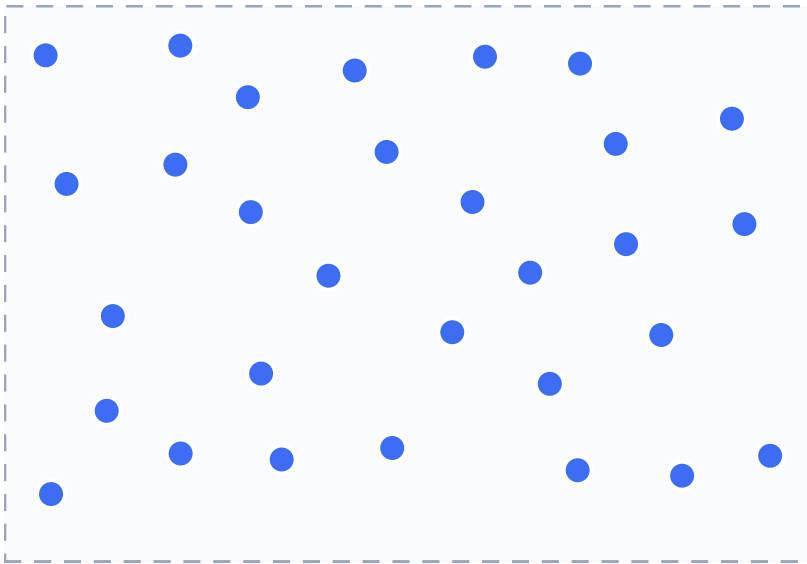
Clustering means sorting data points into groups of similar items. Nobody tells the computer the groups in advance. It has to find them by itself.

Examples: grouping songs by style, or finding busy areas on a map.

What is density?

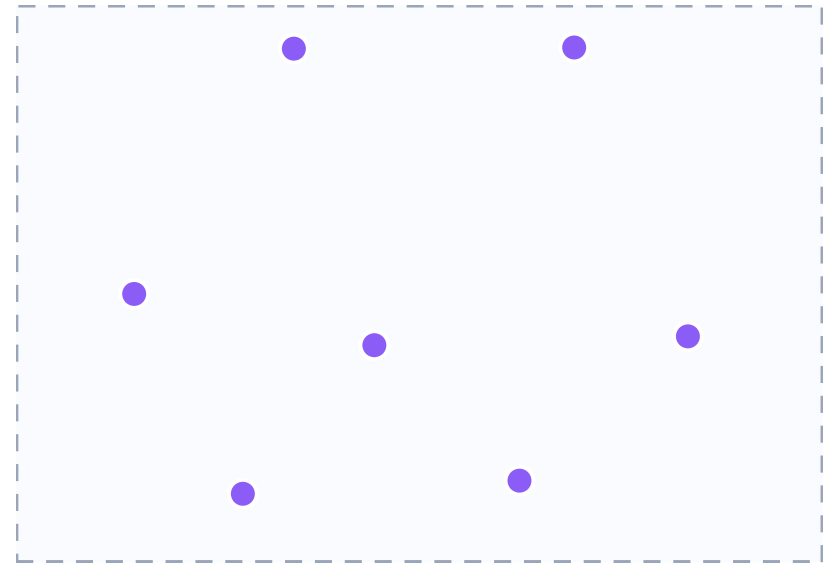
High density

Many points in the box



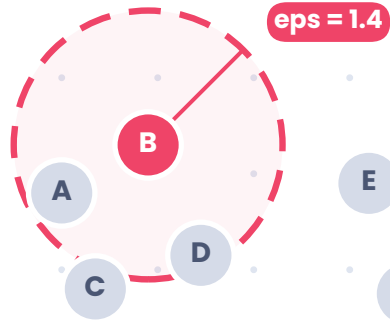
Low density

Few points in the same box



Density means how many points are packed into the same amount of space. Real data often has crowded places and empty places at the same time, like a city with a busy downtown and quiet suburbs.

A quick look at DBSCAN



Draw a circle

DBSCAN draws a circle of radius eps around a point. Here $\text{eps} = 1.4$. It also needs MinPts , the number of points that counts as a crowd. Here $\text{MinPts} = 3$.

Vocabulary

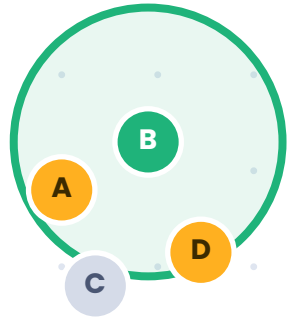
eps the radius of the circle

MinPts points needed to count as a crowd

core point has at least MinPts points in its circle

noise belongs to no cluster

DBSCAN: count the points



3 points inside $\geq$ MinPts

B is a core point

Count the points inside

B's circle holds 3 points (A, B and D). That is at least MinPts, so B is a core point. A core point sits in a crowded spot.

Vocabulary

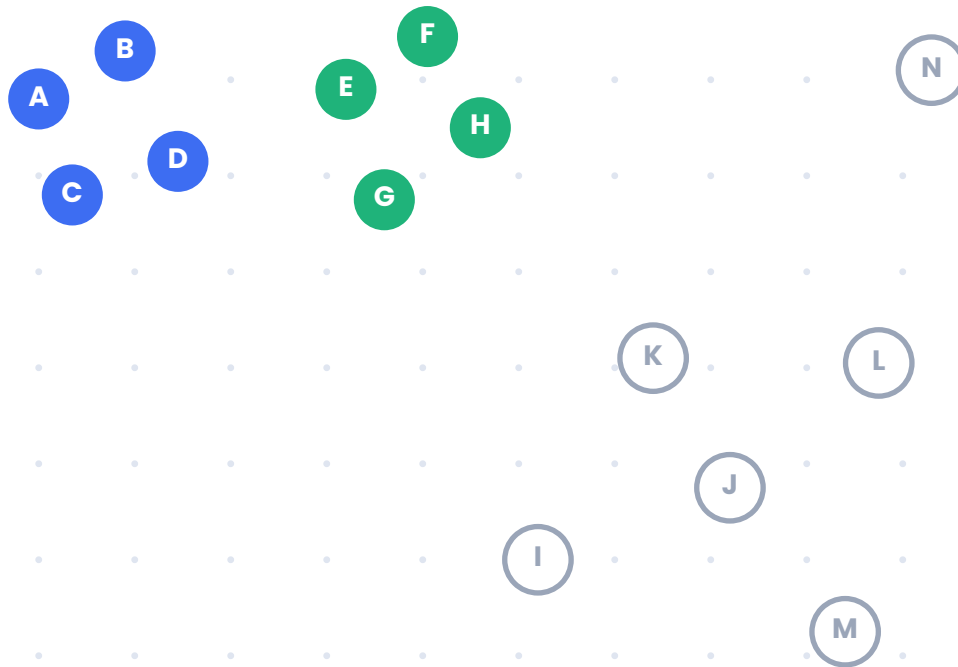
eps the radius of the circle

MinPts points needed to count as a crowd

core point has at least MinPts points in its circle

noise belongs to no cluster

DBSCAN: clusters grow



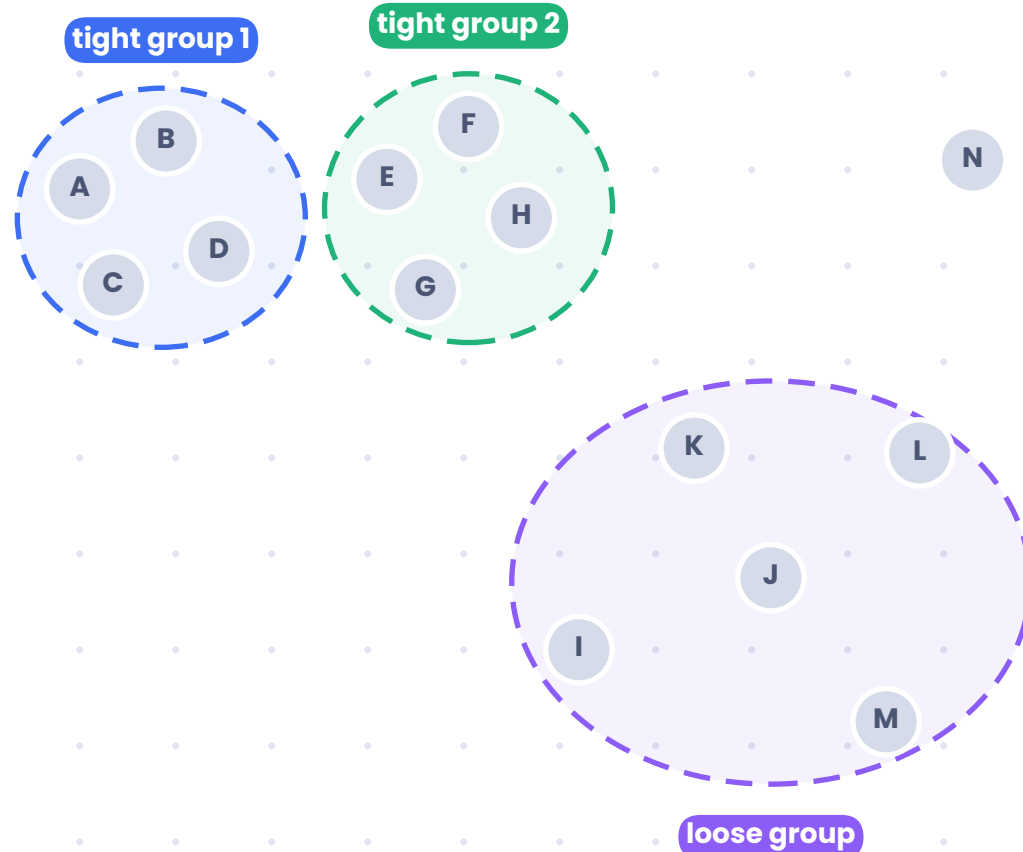
Clusters grow

Core points pull their neighbors into the same cluster, and the cluster keeps spreading. Points that no core point can reach are noise (hollow dots).

Vocabulary

eps	the radius of the circle
MinPts	points needed to count as a crowd
core point	has at least MinPts points in its circle
noise	belongs to no cluster

The data has different densities



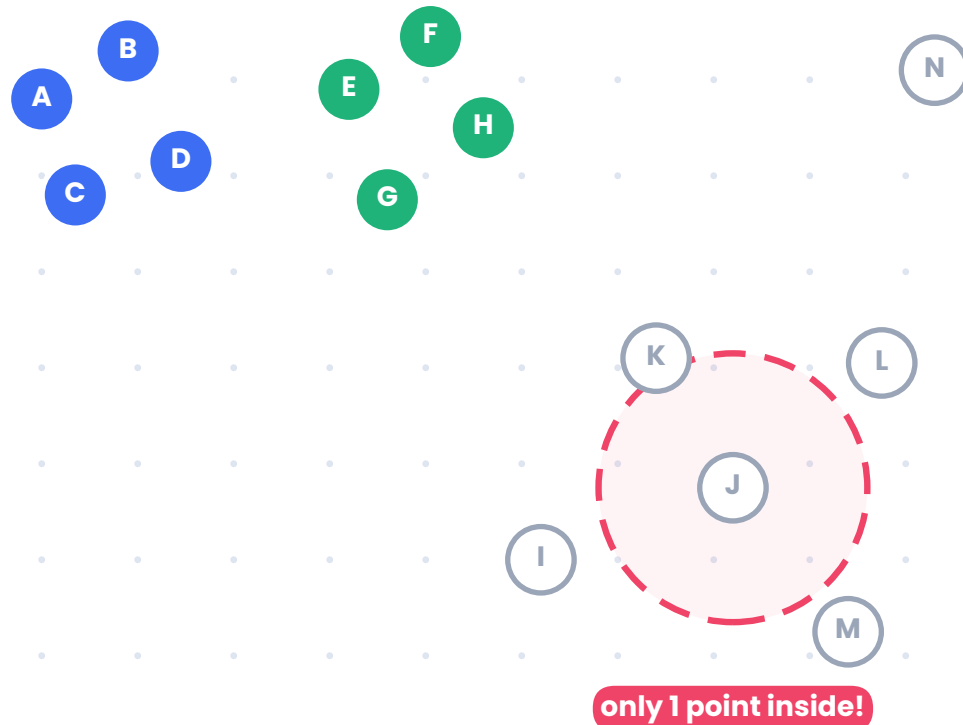
Meet our data

Two groups are tightly packed. One group is spread out. One point sits all alone. We want an algorithm that finds all three groups.

Did we find all 3 groups?

- | | |
|-----------------|------------------|
| ● Tight group 1 | ⓪ not tested yet |
| ● Tight group 2 | ⓪ not tested yet |
| ● Loose group | ⓪ not tested yet |

Try a small eps



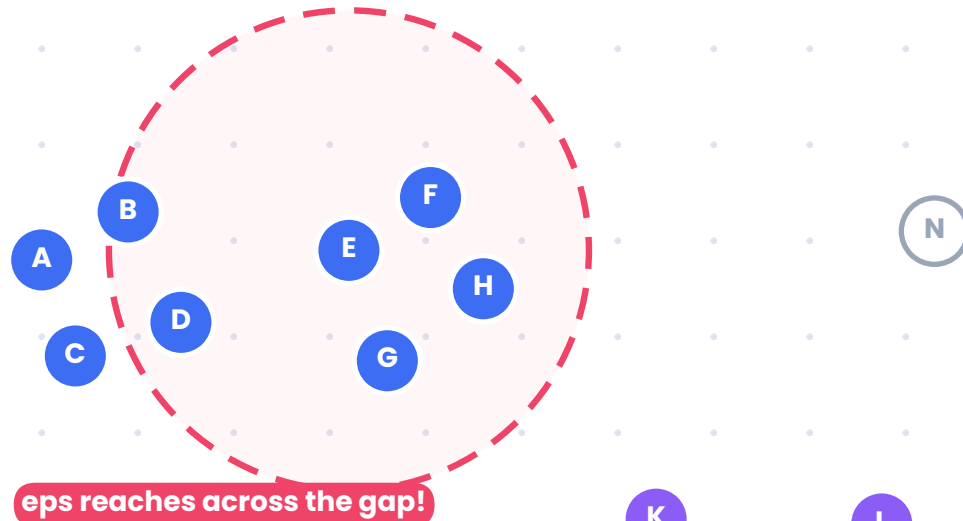
Small eps = 1.4

The tight groups are found. But in the loose group, no circle holds enough points, so all of it is called noise (hollow dots).

Did we find all 3 groups?

- | | |
|-----------------|-----------------|
| ● Tight group 1 | ✓ found |
| ● Tight group 2 | ✓ found |
| ● Loose group | ✗ lost as noise |

Try a big eps



Big eps = 2.5

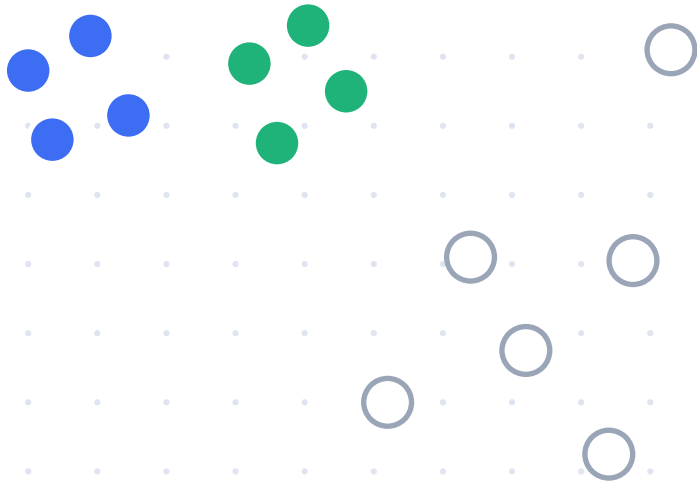
The loose group is found. But eps is now bigger than the gap between the two tight groups, so they get glued into one cluster.

Did we find all 3 groups?

- | | |
|-----------------|--------------------|
| ● Tight group 1 | ✗ glued to group 2 |
| ● Tight group 2 | ✗ glued to group 1 |
| ● Loose group | ✓ found |

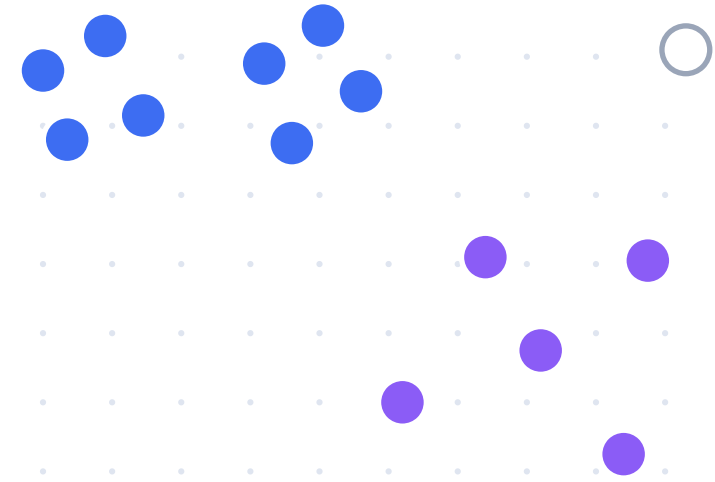
No single eps works

eps = 1.4



The loose group is lost

eps = 2.5



The tight groups merge

DBSCAN uses one eps for all the data. When the densities differ, no single eps finds all three groups.

OPTICS: do not pick one radius

1

Visit the points in a smart order.

We always move to a crowded neighbor first.

2

Write down one number per point.

The reachability: how far we had to go to reach it.

3

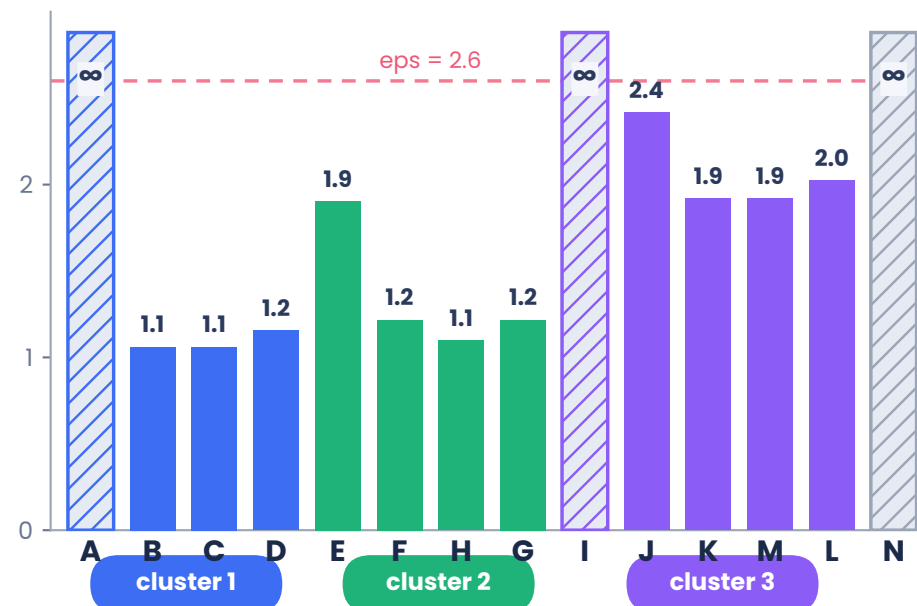
Draw the numbers as bars.

Valleys are clusters. Peaks are gaps.

Think of a walk

Walk from house to house and write down each walk's length. Short walks mean a busy neighborhood. A long walk means you crossed empty land.

The result: a reachability plot



Low bars = crowded. Tall bars = gaps.

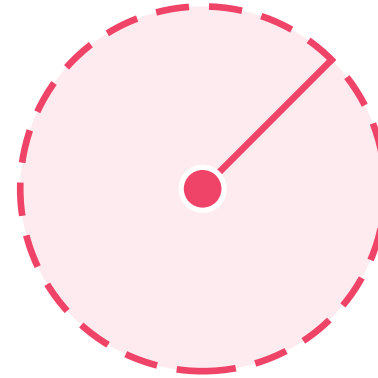
OPTICS was introduced in 1999 by Ankerst, Breunig, Kriegel and Sander.

Two settings



MinPts = 3

How many points make a crowd, counting the point itself.



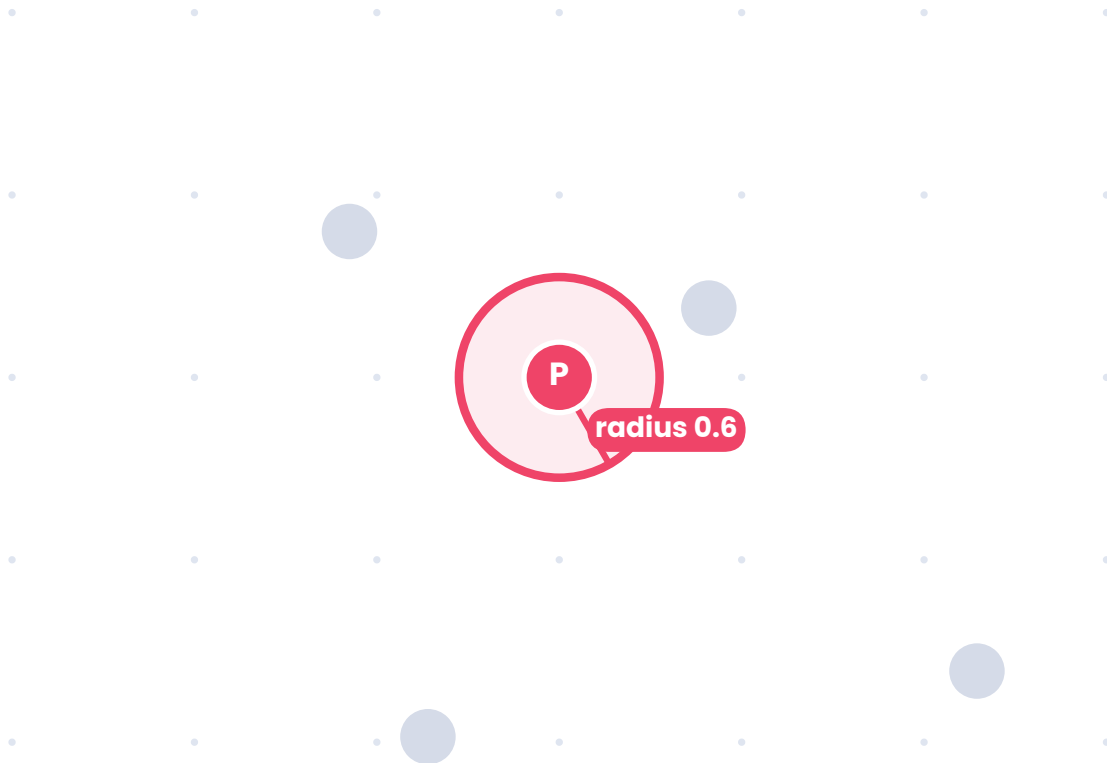
eps = 2.6

The biggest radius we are willing to look at.

A big difference from DBSCAN

In DBSCAN, eps decides the clusters. In OPTICS, eps is only an upper limit that keeps the work small. The plot we build afterwards works for every radius up to eps.

Core distance: start small



Start small

Take a point P and draw a tiny circle around it. Only P is inside, so we count 1 point. We need 3, which is MinPts.

Points inside the circle

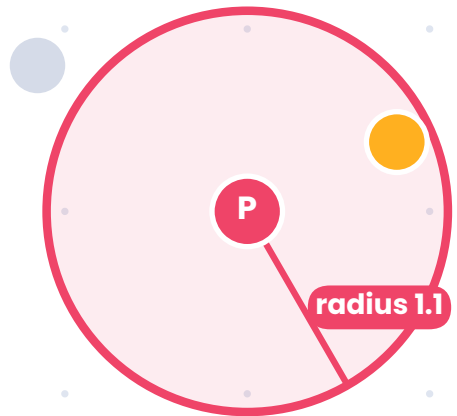


1 of 3

Core distance

the smallest radius that holds MinPts points, counting the point itself.

Core distance: grow the circle



Grow the circle

We make the circle bigger. It catches one more point, so 2 points are inside. That is still fewer than 3, so we keep growing.

Points inside the circle

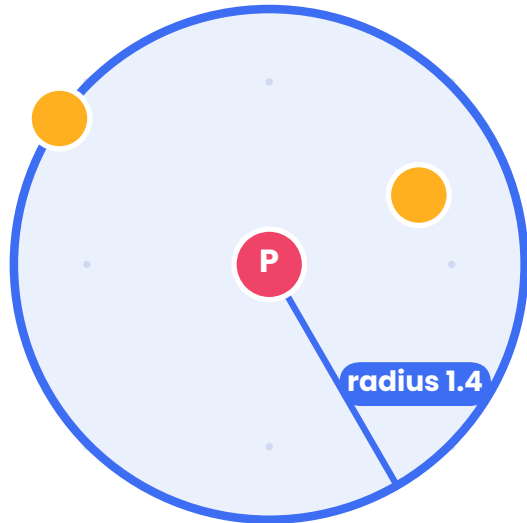


2 of 3

Core distance

the smallest radius that holds MinPts points, counting the point itself.

Core distance: stop at 3 points



Stop at 3 points

Now 3 points are inside: P and two others. We stop here. The radius of this circle, 1.4, is the core distance of P.

Points inside the circle



3 of 3

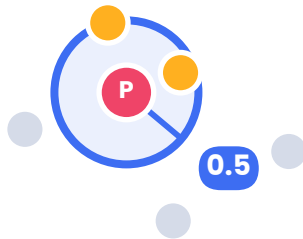
Core distance

the smallest radius that holds MinPts points, counting the point itself.

Crowded or empty?

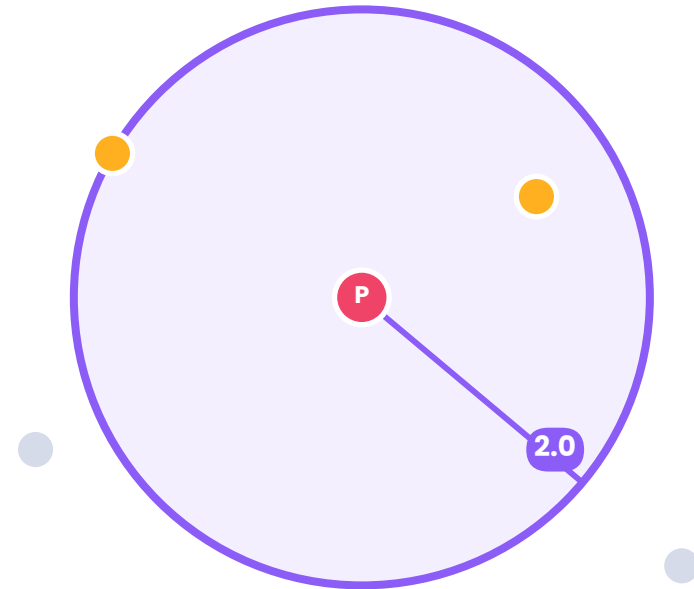
A crowded area

small core distance



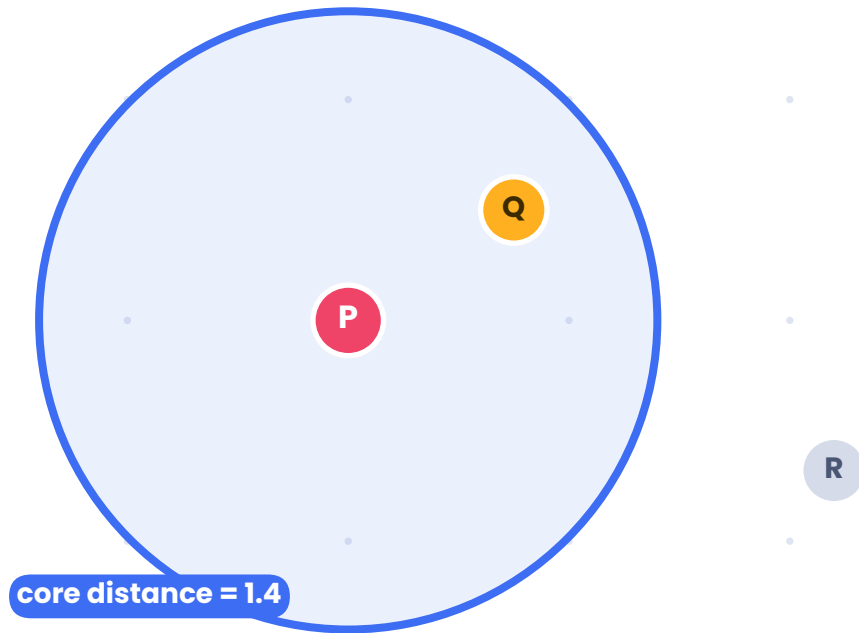
An empty area

large core distance



In a crowded area a small circle is enough, so the core distance is small. In an empty area we need a big circle, so the core distance is large.

Reachability distance



The larger of two numbers

How hard is it to reach a point from P? We take the larger of two numbers: the core distance of P, and the distance from P to the point.

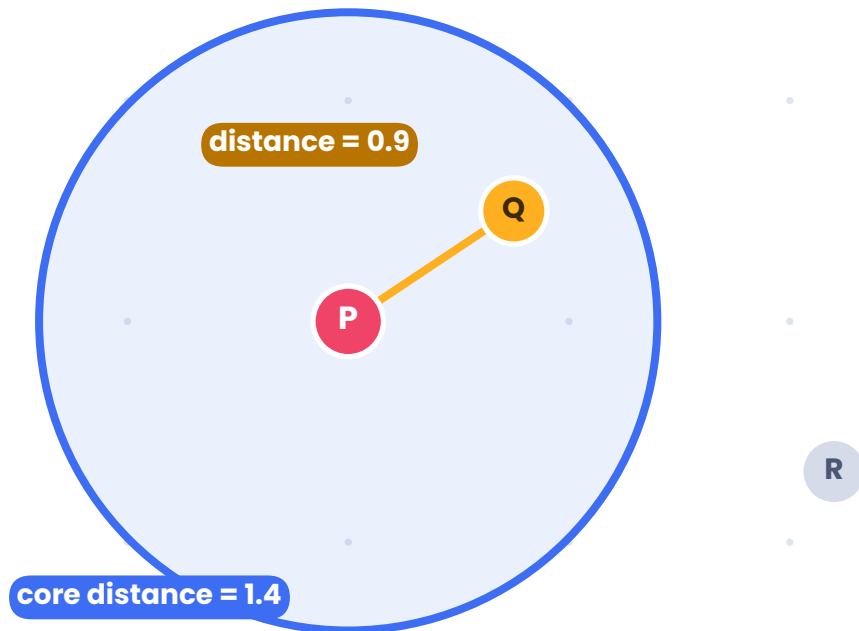
Formula

$$\text{reach} = \max(\text{core distance}, \text{distance})$$

The bigger number wins.

Small number = easy to reach (a crowded area).
Big number = hard to reach (a gap).

Reachability: a close neighbor



Q is inside the circle

Q is only 0.9 away, which is less than the core distance 1.4. Being this close does not make it easier than the core distance. So Q's reachability is 1.4.

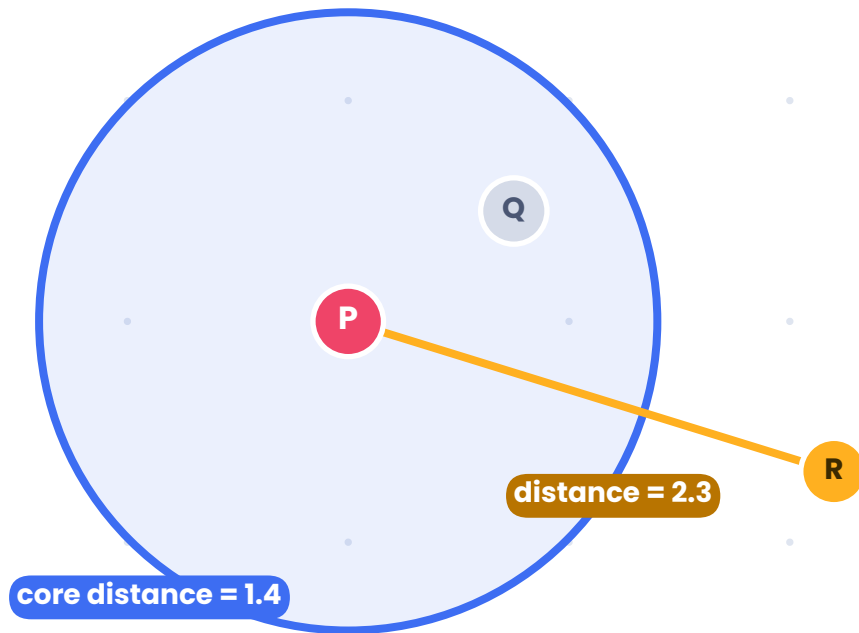
Formula

$\text{reach} = \max(\text{core distance}, \text{distance})$

$\max(1.4, 0.9) = 1.4$

Small number = easy to reach (a crowded area).
Big number = hard to reach (a gap).

Reachability: a far neighbor



R is outside the circle

R is 2.3 away, which is more than the core distance 1.4. So R's reachability is its real distance, 2.3. A big number means R is hard to reach.

Formula

$\text{reach} = \max(\text{core distance}, \text{distance})$

$\max(1.4, 2.3) = 2.3$

Small number = easy to reach (a crowded area).
Big number = hard to reach (a gap).

OPTICS in five steps

1

Pick

Pick a point that is not visited. If the waiting list has points, take the one with the smallest number.

2

Measure

Find its core distance: the smallest circle (up to ϵ) that holds MinPts points.

3

Update

For each unvisited neighbor inside ϵ , work out its reachability. Keep the smaller number.

4

Move

Add the new neighbors to the waiting list. Go to the waiting-list point with the smallest number.

5

Repeat

Stop when every point is visited. The bars, in visiting order, make the reachability plot.

Words to remember**Core distance**

The smallest radius that holds MinPts points.

Reachability distance

$\max(\text{core distance, distance to the neighbor})$.

Waiting list

Points we have seen but not visited yet. The smallest number goes first.

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Reachability distance

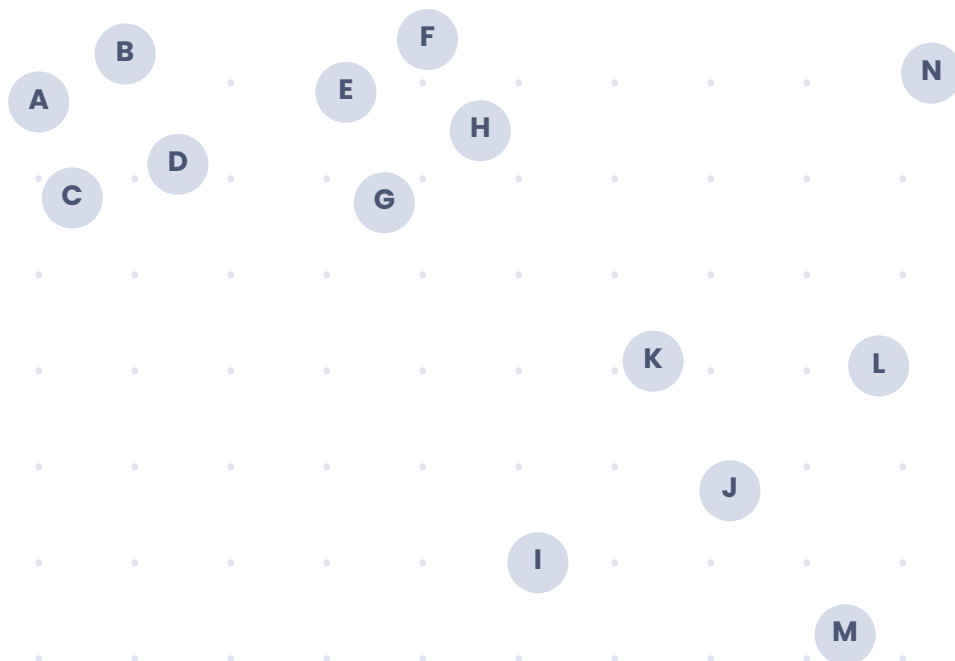
$\max(\text{core distance, distance to the neighbor})$.

Waiting list

Points we have seen but not visited yet. The smallest number goes first.

Our data: 14 points

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

Let's run OPTICS by hand

We use the 14 points on the left, named A to N. Settings: MinPts = 3 and eps = 2.6. We start at point A.

Watch these three things

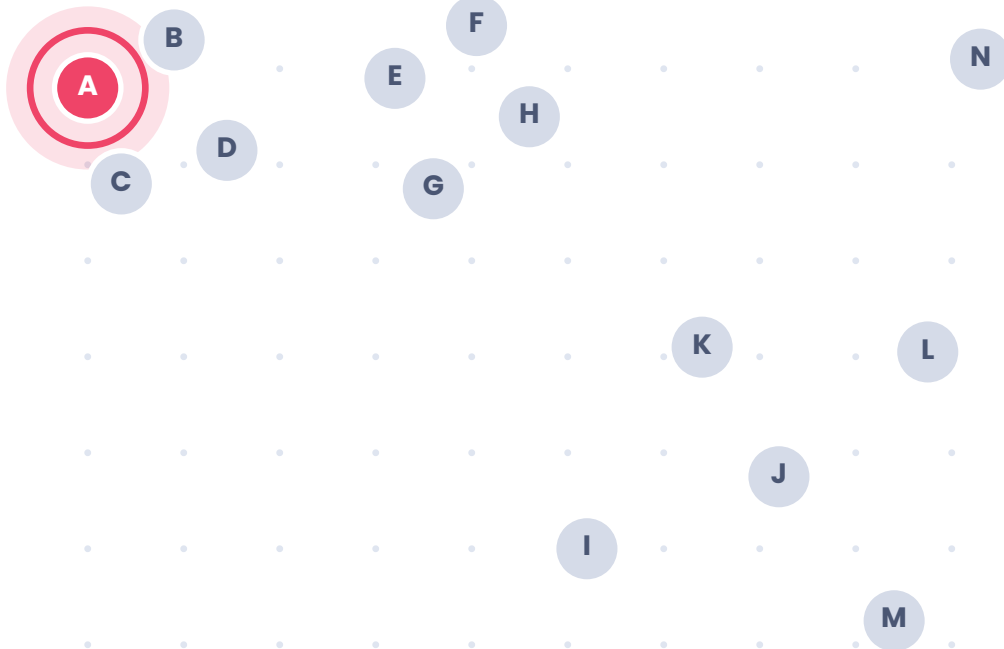
- 1 The circles around the current point
- 2 The waiting list of points to visit next
- 3 The plot that fills up, bar by bar

Visit point A

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

Visit A

The waiting list is empty, so we simply start with A. It has no previous point, so its reachability is undefined (∞). We add a full-height bar.

Waiting list (smallest goes first)

Empty. Start fresh!

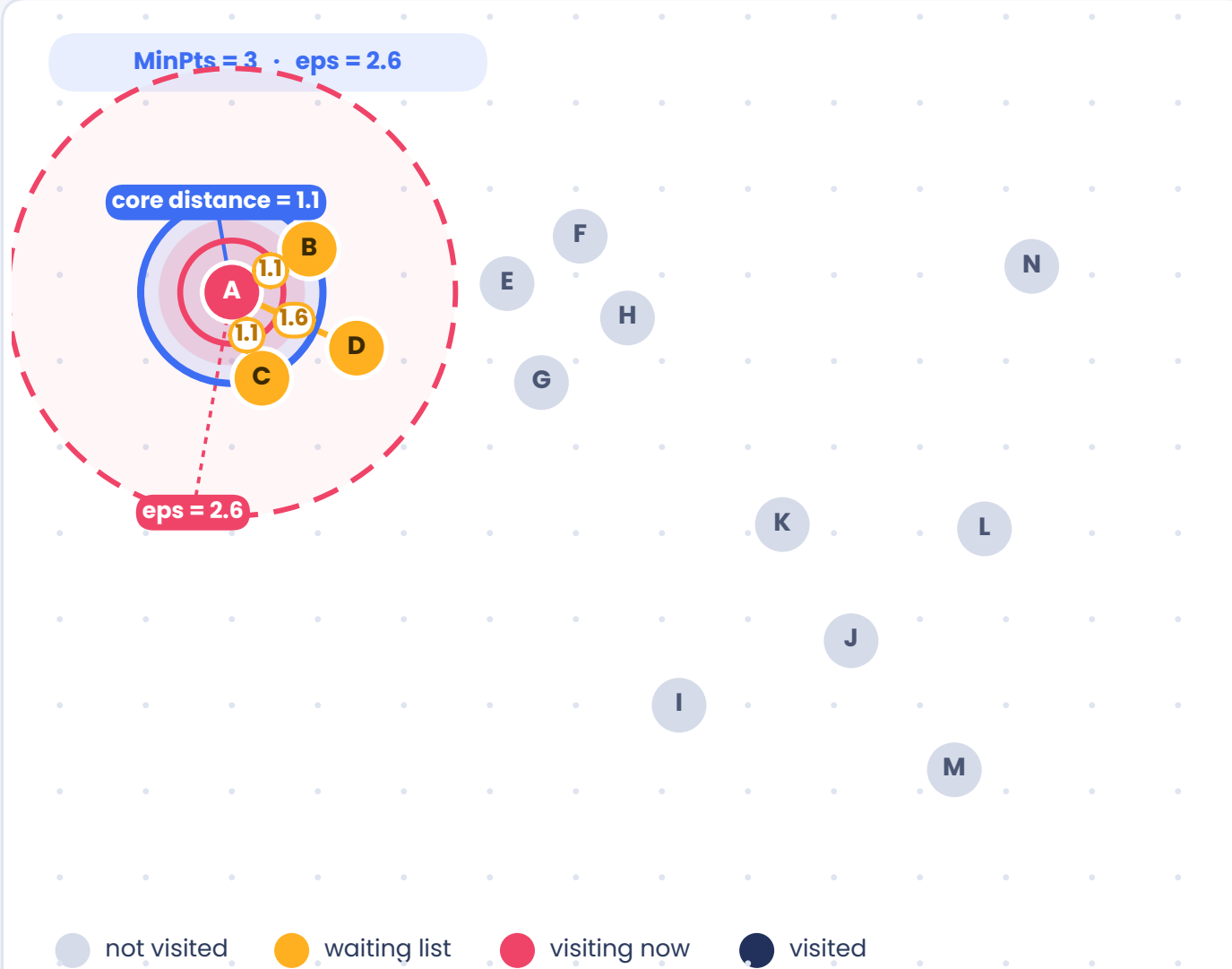
Reachability plot (bars appear in visiting order)



Look around point A

1 Visit

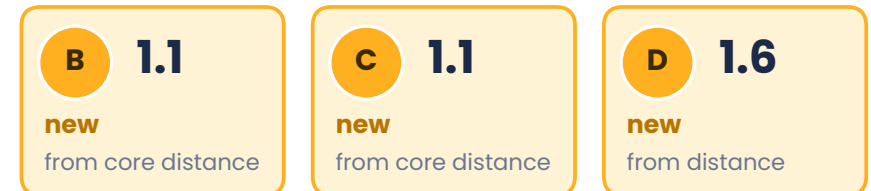
2 Look around



A is a core point

4 points fit inside eps (counting A). That is at least MinPts = 3. Core distance = 1.1. Each unvisited neighbor gets $\max(\text{core distance, distance to A})$. We keep the smaller value.

Waiting list after the update



Reachability plot (bars appear in visiting order)

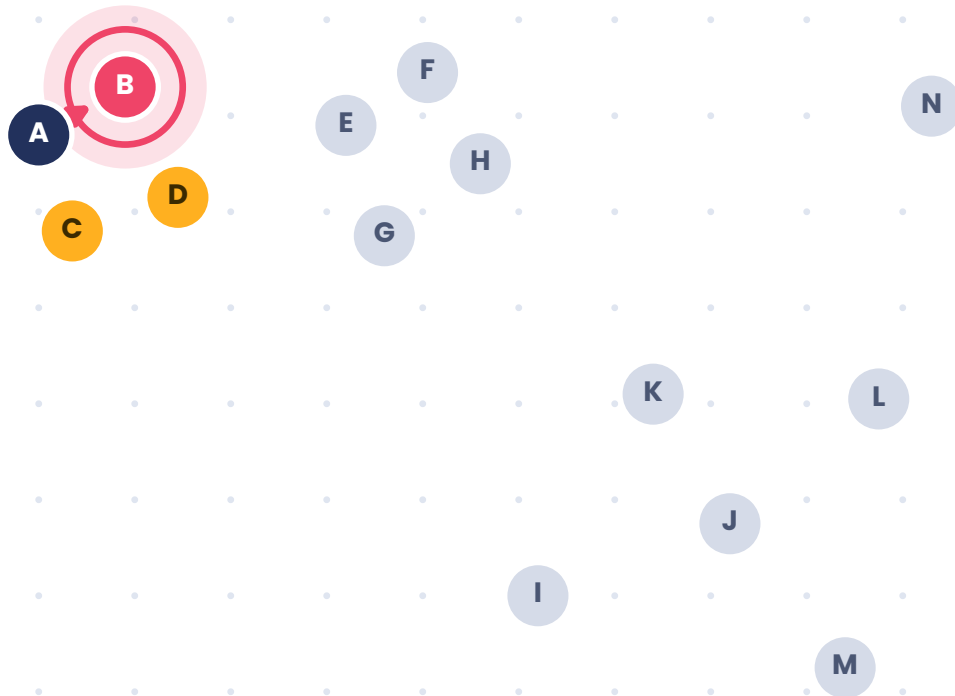


Visit point B

1 Visit

2 Look around

MinPts = 3 · eps = 2.6

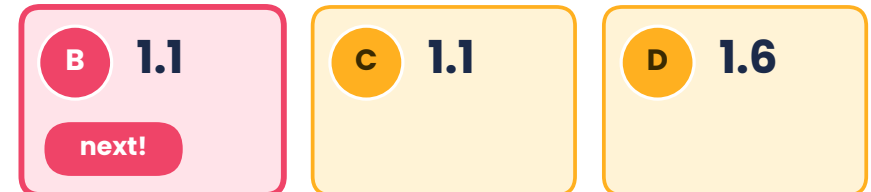


● not visited
 ● waiting list
 ● visiting now
 ● visited

Visit B

B has the smallest number in the waiting list (1.1), so we visit it next. We add a bar of height 1.1 to the plot. (Tie? We go alphabetically.)

Waiting list (smallest goes first)



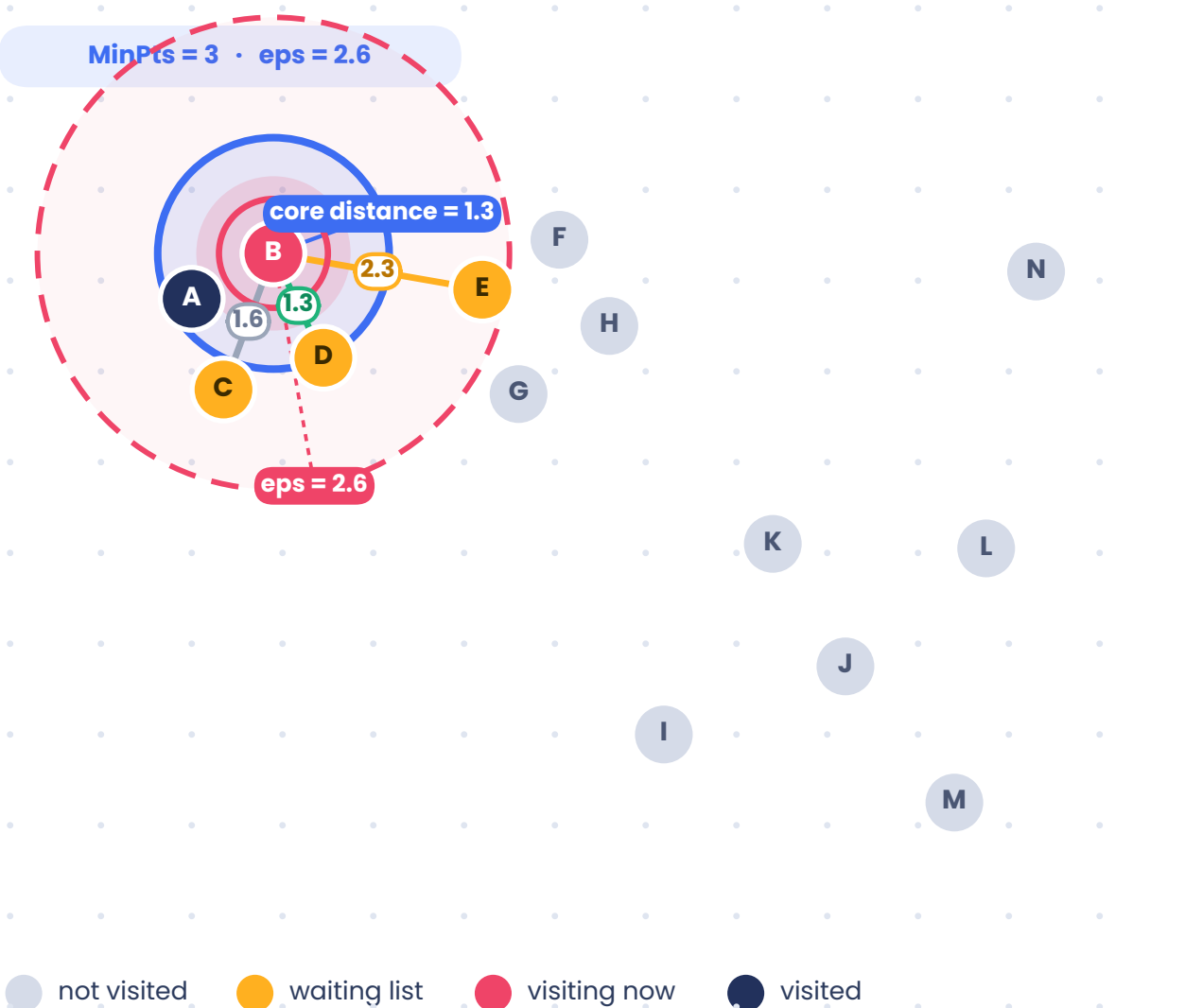
Reachability plot (bars appear in visiting order)



Look around point B

1 Visit

2 Look around



B is a core point

5 points fit inside eps (counting B). That is at least MinPts = 3. Core distance = 1.3. Each unvisited neighbor gets $\max(\text{core distance, distance to B})$. We keep the smaller value.

Waiting list after the update

C 1.1

kept (offer 1.6)
bigger, so ignored

D 1.3

lower (was 1.6)
from core distance

E 2.3

new
from distance

Reachability plot (bars appear in visiting order)

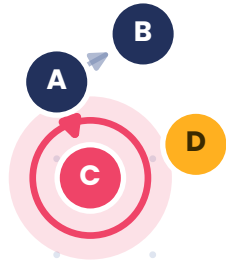


Visit point C

1 Visit

2 Look around

MinPts = 3 · eps = 2.6

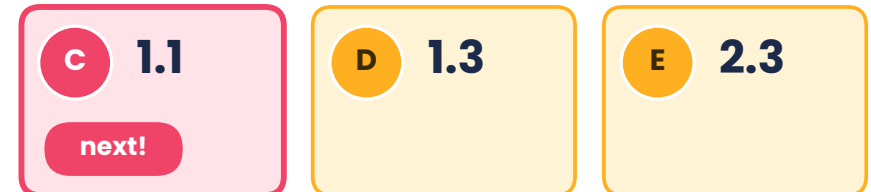


● not visited ● waiting list ● visiting now ● visited

Visit C

C has the smallest number in the waiting list (1.1), so we visit it next. We add a bar of height 1.1 to the plot.

Waiting list (smallest goes first)



Reachability plot (bars appear in visiting order)

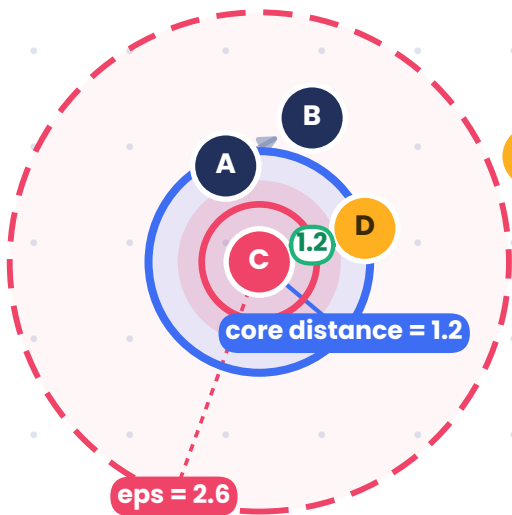


Look around point C

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



C is a core point

4 points fit inside eps (counting C). That is at least MinPts = 3. Core distance = 1.2. Each unvisited neighbor gets $\max(\text{core distance, distance to C})$. We keep the smaller value.

Waiting list after the update

D 1.2

lower (was 1.3)
from core distance

E 2.3

Reachability plot (bars appear in visiting order)

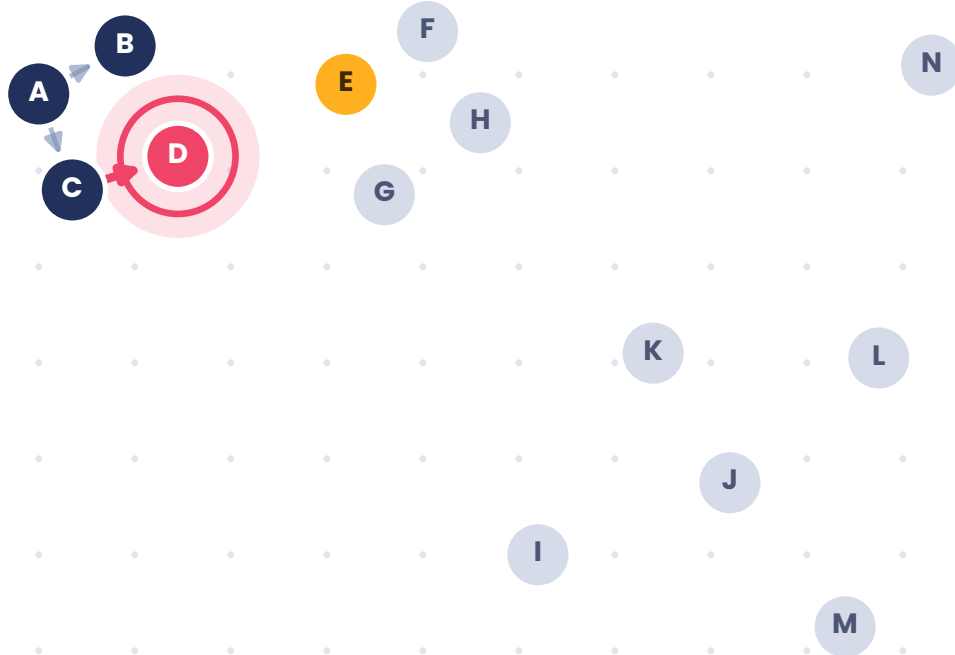


Visit point D

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

Visit D

D has the smallest number in the waiting list (1.2), so we visit it next. We add a bar of height 1.2 to the plot.

Waiting list (smallest goes first)



Reachability plot (bars appear in visiting order)

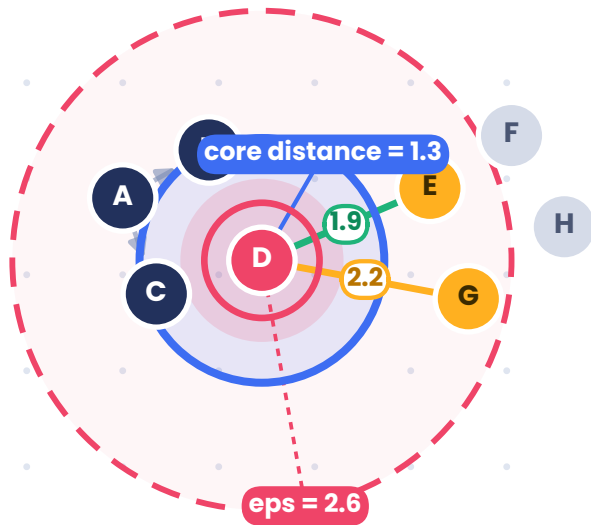


Look around point D

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

D is a core point

6 points fit inside eps (counting D). That is at least MinPts = 3. Core distance = 1.3. Each unvisited neighbor gets $\max(\text{core distance, distance to D})$. We keep the smaller value.

Waiting list after the update

E 1.9

lower (was 2.3)
from distance

G 2.2

new
from distance

Reachability plot (bars appear in visiting order)

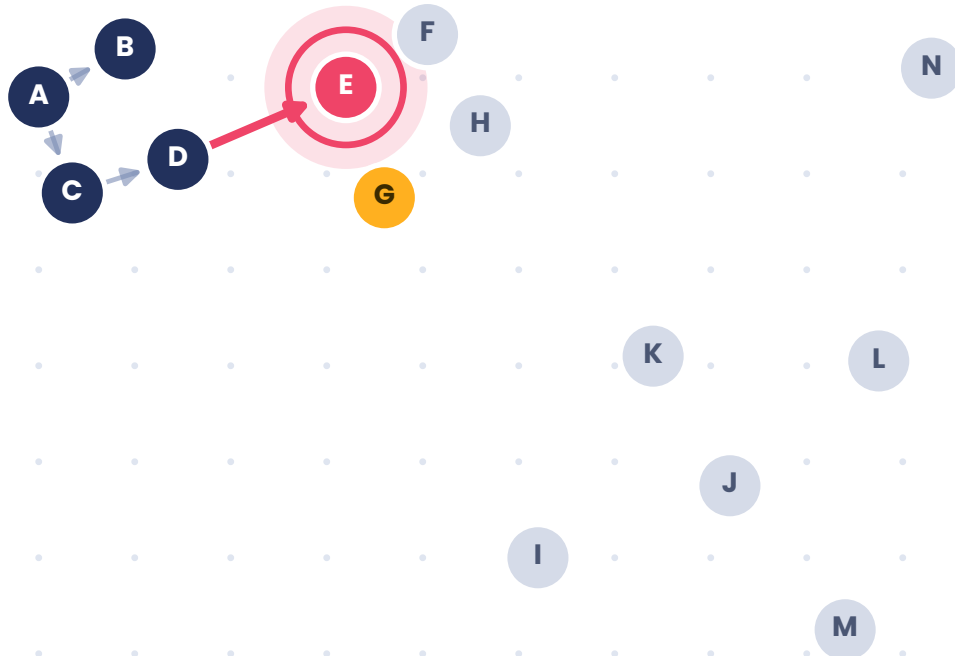


Visit point E

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

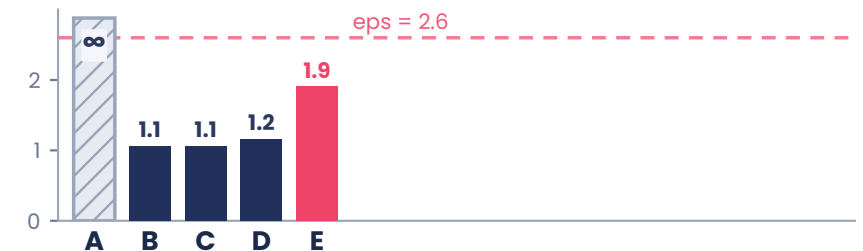
Visit E

E has the smallest number in the waiting list (1.9), so we visit it next. We add a bar of height 1.9 to the plot.

Waiting list (smallest goes first)



Reachability plot (bars appear in visiting order)

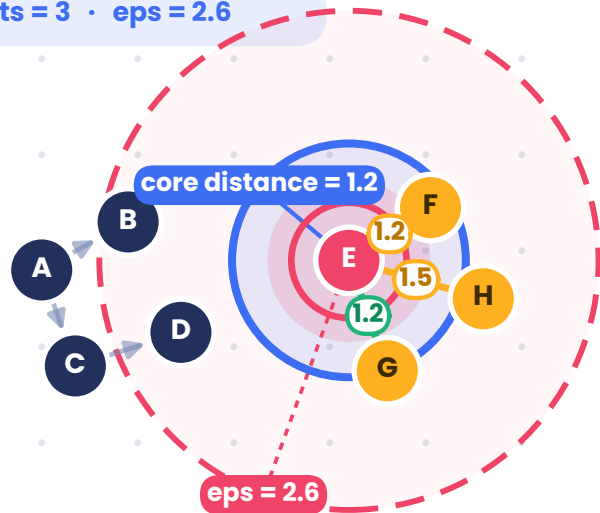


Look around point E

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

E is a core point

6 points fit inside eps (counting E). That is at least MinPts = 3. Core distance = 1.2. Each unvisited neighbor gets $\max(\text{core distance, distance to E})$. We keep the smaller value.

Waiting list after the update

F 1.2

new
from core distance

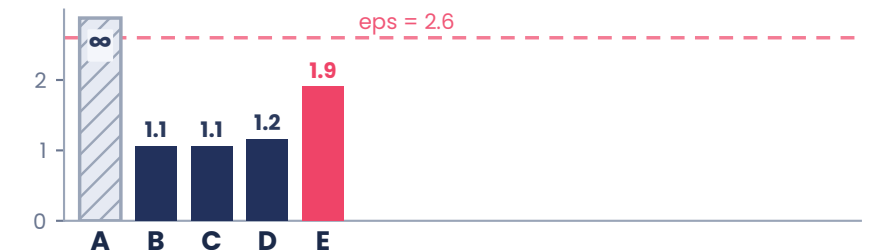
G 1.2

lower (was 2.2)
from core distance

H 1.5

new
from distance

Reachability plot (bars appear in visiting order)

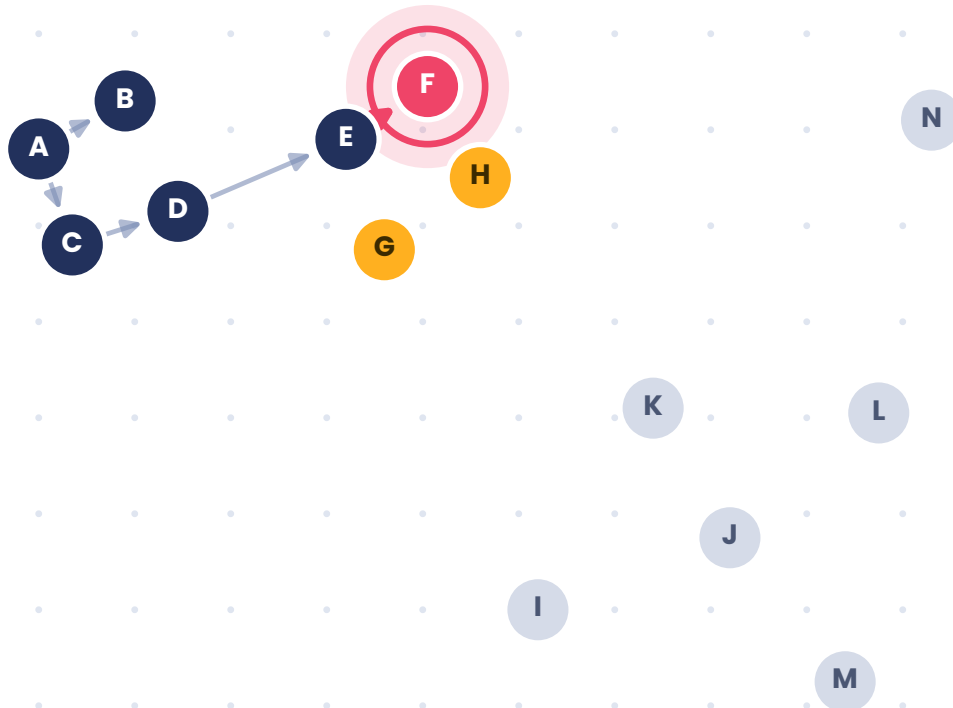


Visit point F

1 Visit

2 Look around

MinPts = 3 · eps = 2.6

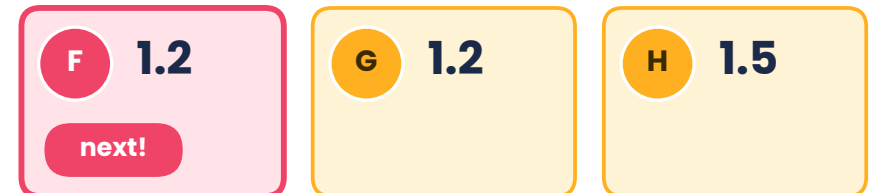


● not visited ● waiting list ● visiting now ● visited

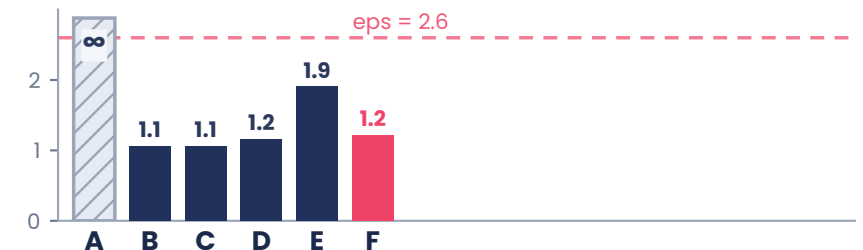
Visit F

F has the smallest number in the waiting list (1.2), so we visit it next. We add a bar of height 1.2 to the plot. (Tie? We go alphabetically.)

Waiting list (smallest goes first)



Reachability plot (bars appear in visiting order)

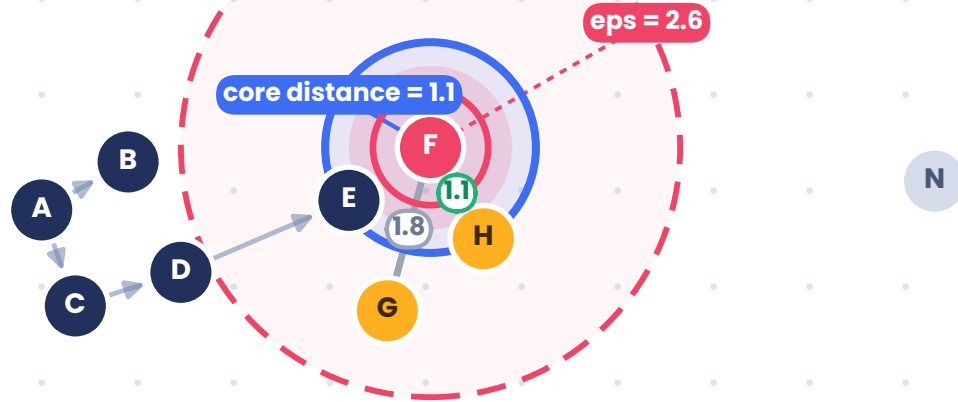


Look around point F

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



F is a core point

4 points fit inside eps (counting F). That is at least MinPts = 3. Core distance = 1.1. Each unvisited neighbor gets $\max(\text{core distance, distance to F})$. We keep the smaller value.

Waiting list after the update

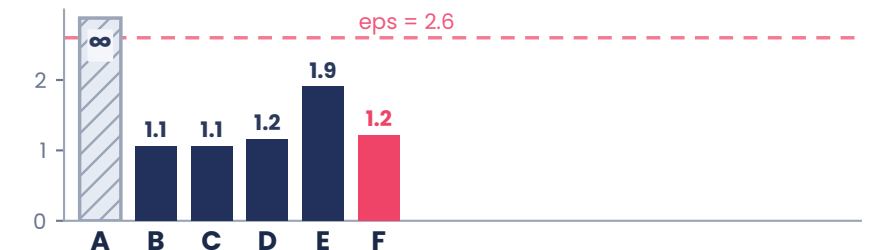
H 1.1

lower (was 1.5)
from core distance

G 1.2

kept (offer 1.8)
bigger, so ignored

Reachability plot (bars appear in visiting order)



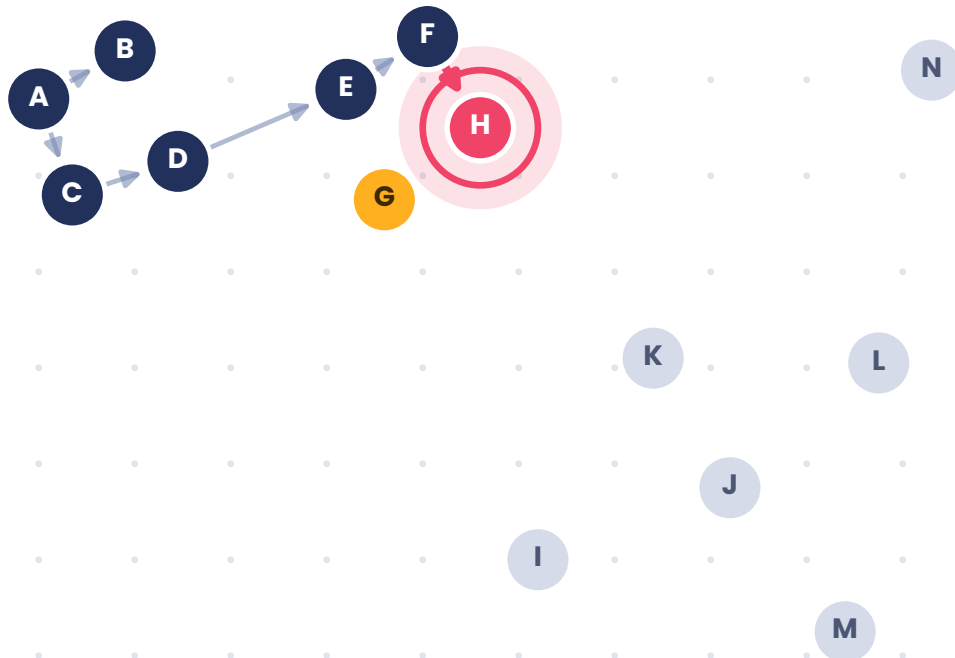
not visited waiting list visiting now visited

Visit point H

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

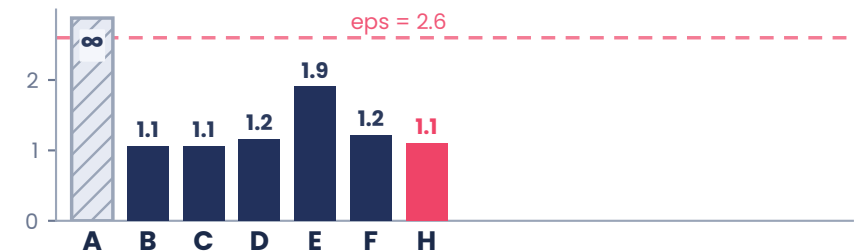
Visit H

H has the smallest number in the waiting list (1.1), so we visit it next. We add a bar of height 1.1 to the plot.

Waiting list (smallest goes first)



Reachability plot (bars appear in visiting order)

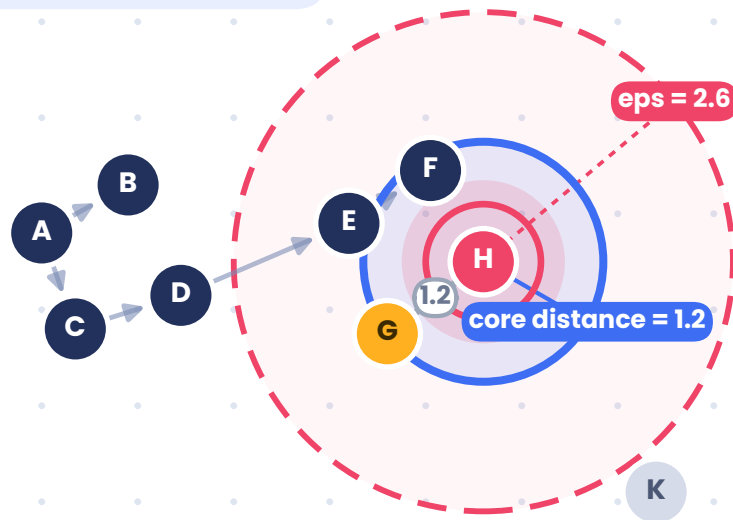


Look around point H

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

H is a core point

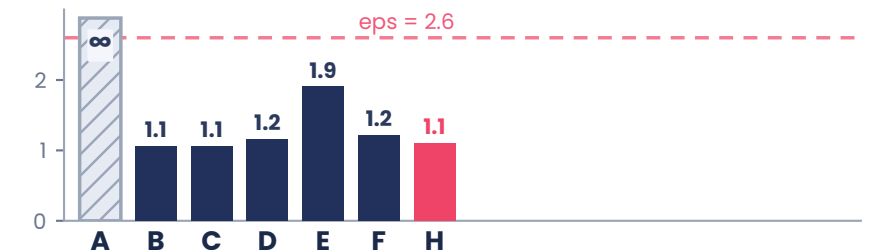
4 points fit inside eps (counting H). That is at least MinPts = 3. Core distance = 1.2. Each unvisited neighbor gets $\max(\text{core distance, distance to H})$. We keep the smaller value.

Waiting list after the update

G 1.2

kept (offer 1.2)
bigger, so ignored

Reachability plot (bars appear in visiting order)

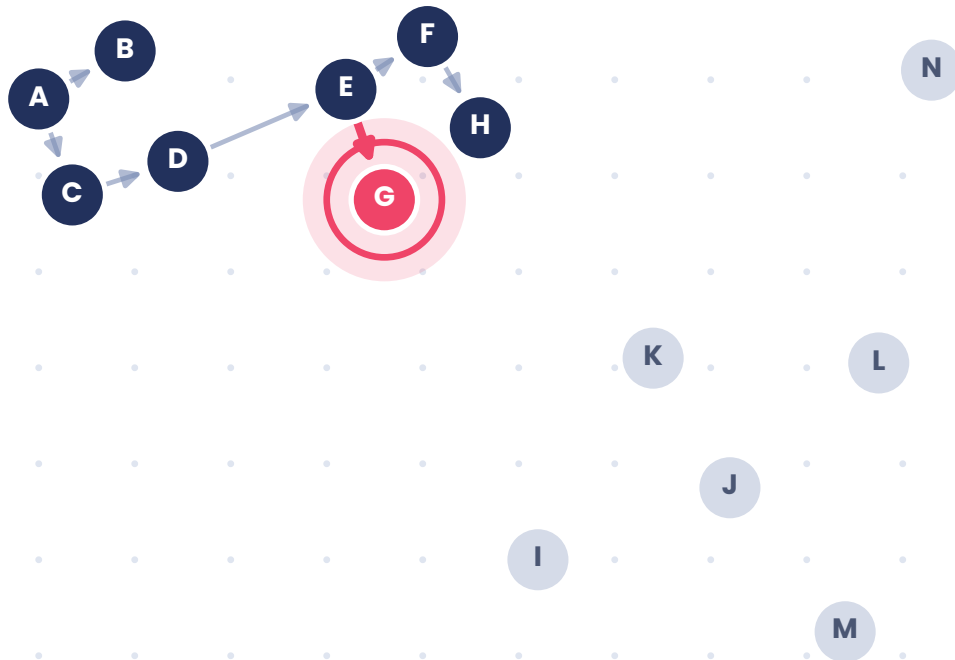


Visit point G

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

Visit G

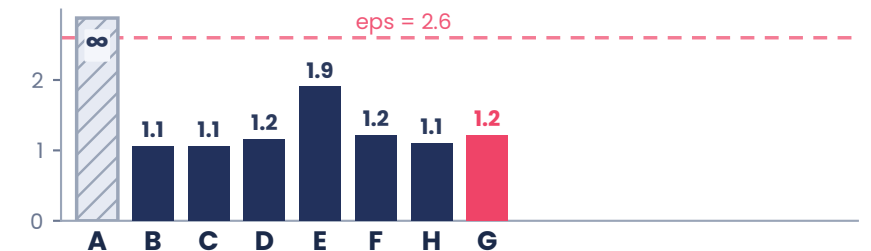
G has the smallest number in the waiting list (1.2), so we visit it next. We add a bar of height 1.2 to the plot.

Waiting list (smallest goes first)

G 1.2

next!

Reachability plot (bars appear in visiting order)

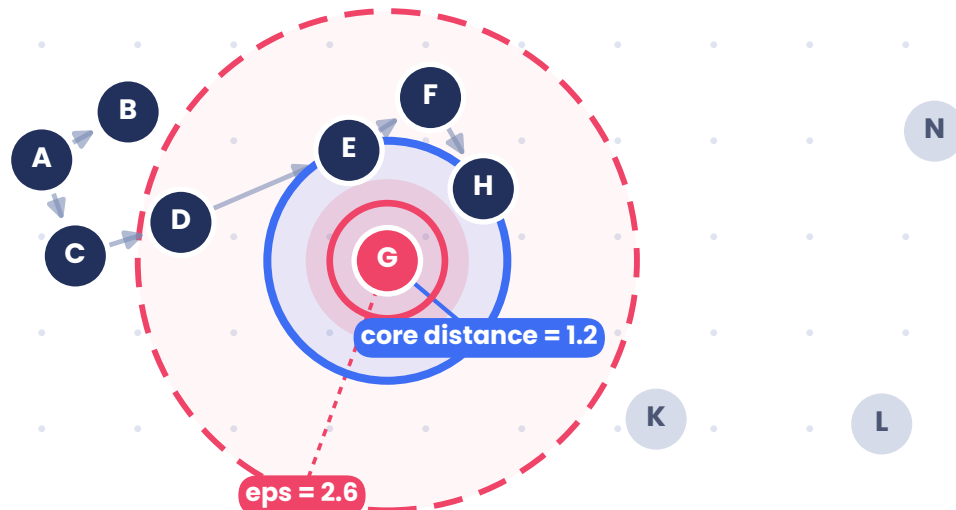


Look around point G

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

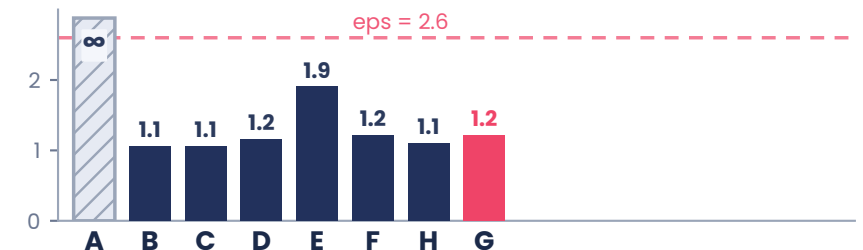
G is a core point

5 points fit inside eps (counting G). That is at least MinPts = 3. Core distance = 1.2. Every neighbor is already visited, so there is nothing to update.

Waiting list after the update

Empty. Nothing is waiting.

Reachability plot (bars appear in visiting order)

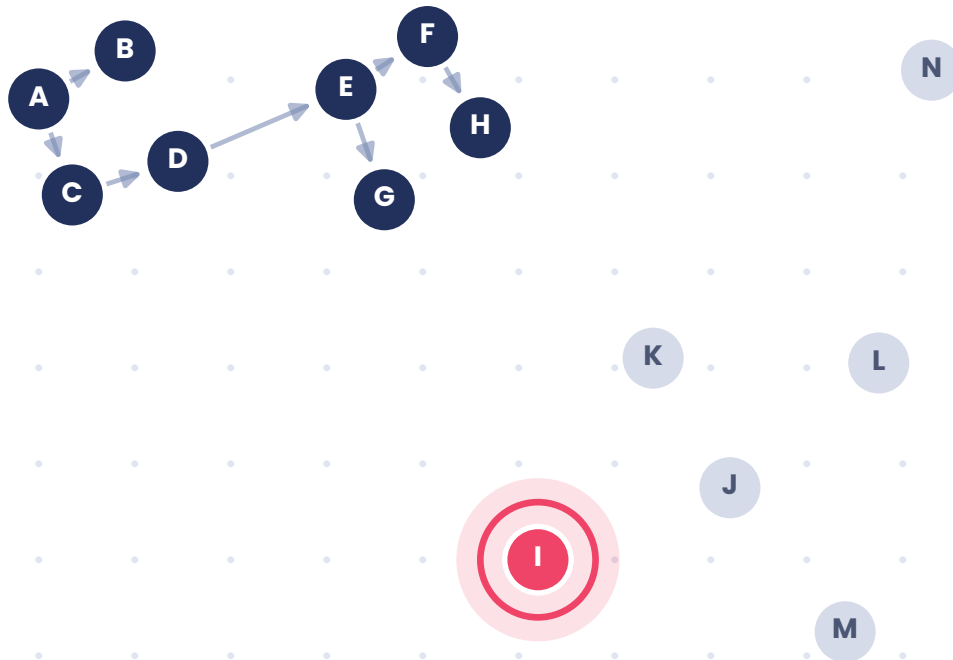


Visit point I

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

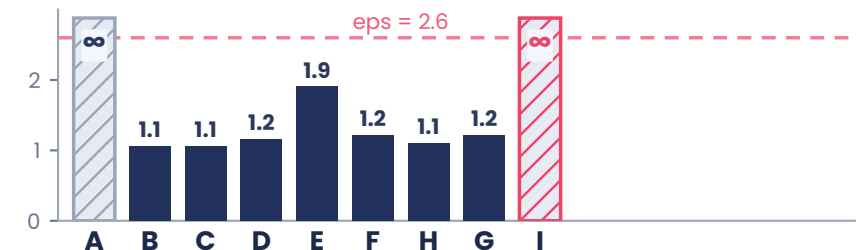
Visit I

The waiting list is empty and nothing is within eps of the last group. So we start fresh at I. Its reachability is undefined (∞).

Waiting list (smallest goes first)

Empty. Start fresh!

Reachability plot (bars appear in visiting order)

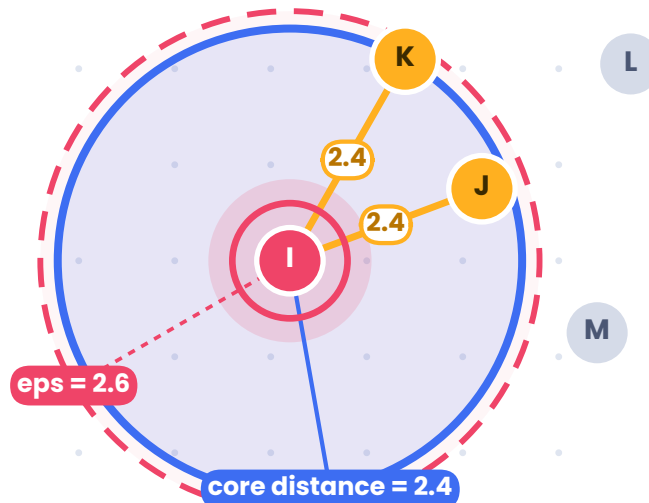
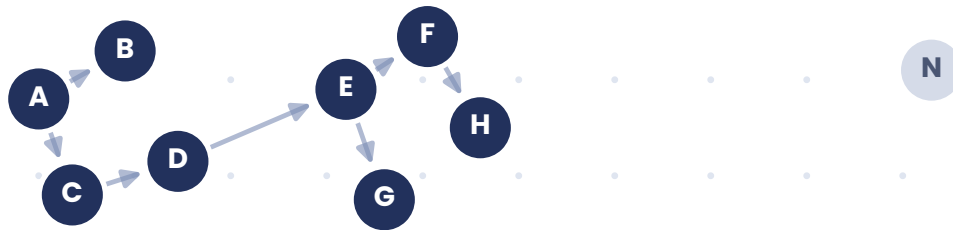


Look around point I

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

I is a core point

3 points fit inside eps (counting I). That is at least MinPts = 3. Core distance = 2.4. Each unvisited neighbor gets $\max(\text{core distance, distance to I})$. We keep the smaller value.

Waiting list after the update

J 2.4

new

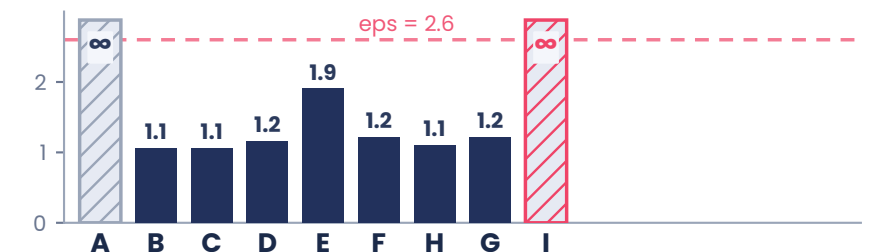
from core distance

K 2.4

new

from core distance

Reachability plot (bars appear in visiting order)

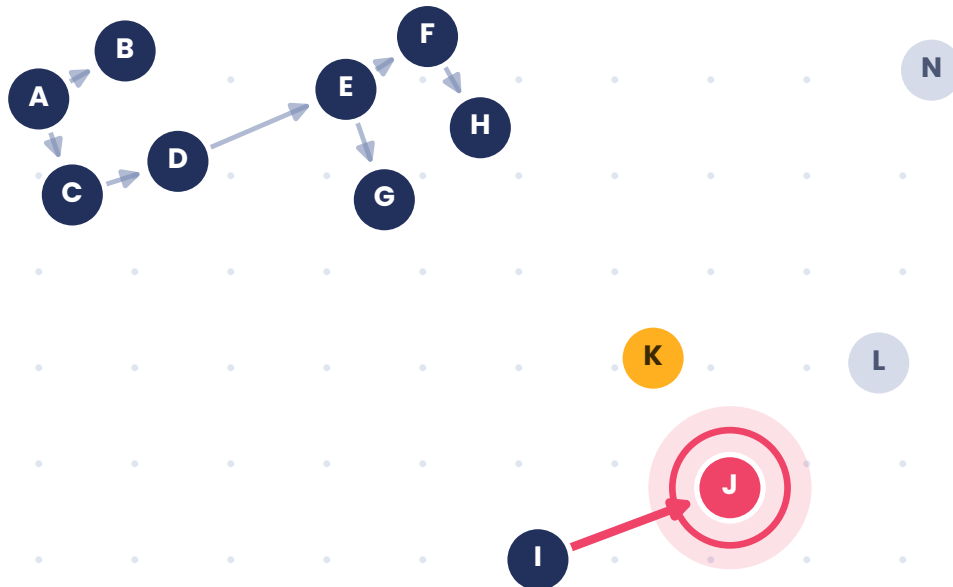


Visit point J

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

Visit J

J has the smallest number in the waiting list (2.4), so we visit it next. We add a bar of height 2.4 to the plot. (Tie? We go alphabetically.)

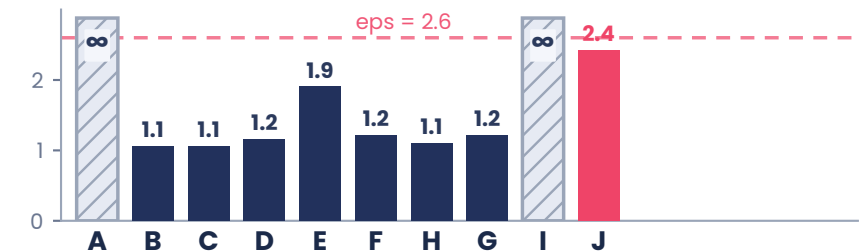
Waiting list (smallest goes first)

J 2.4

next!

K 2.4

Reachability plot (bars appear in visiting order)

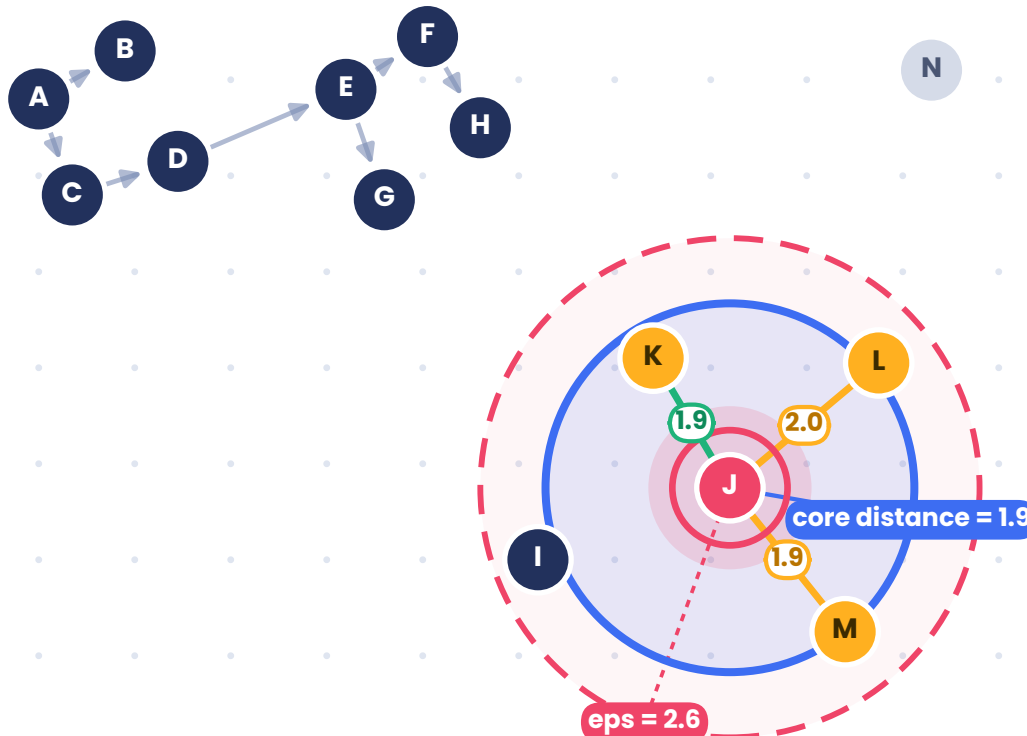


Look around point J

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



not visited waiting list visiting now visited

J is a core point

5 points fit inside eps (counting J). That is at least MinPts = 3. Core distance = 1.9. Each unvisited neighbor gets $\max(\text{core distance, distance to J})$. We keep the smaller value.

Waiting list after the update

K 1.9

lower (was 2.4)
from core distance

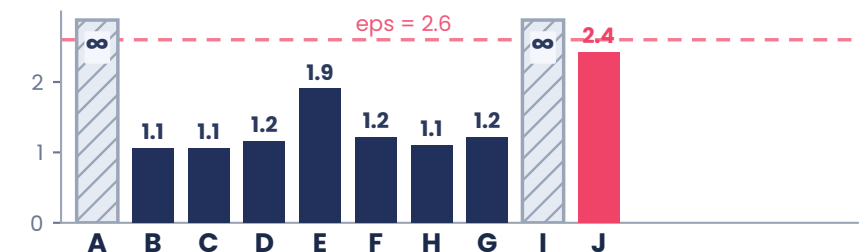
M 1.9

new
from core distance

L 2.0

new
from distance

Reachability plot (bars appear in visiting order)

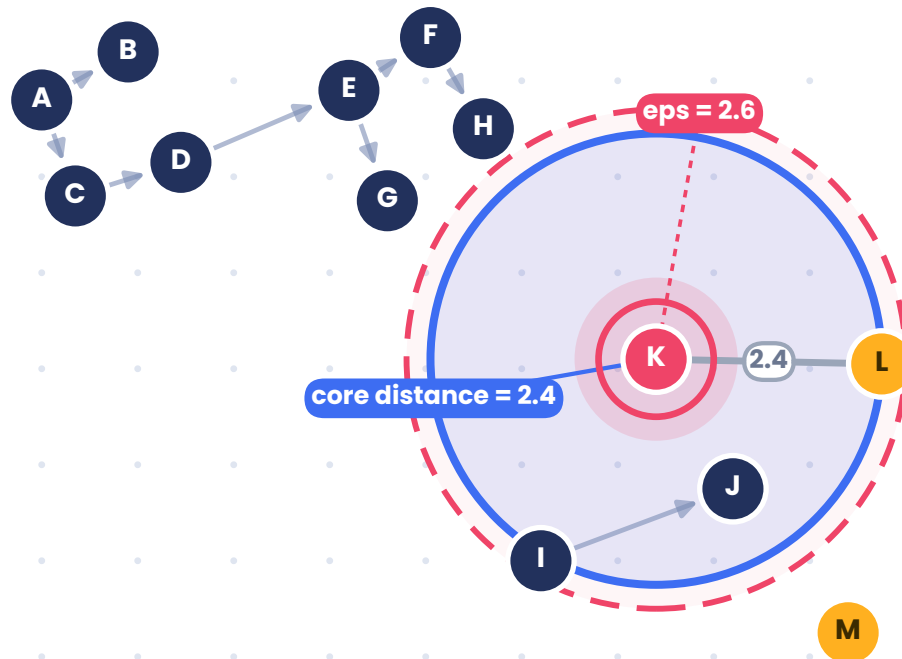


Look around point K

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

K is a core point

4 points fit inside eps (counting K). That is at least MinPts = 3. Core distance = 2.4. Each unvisited neighbor gets $\max(\text{core distance, distance to K})$. We keep the smaller value.

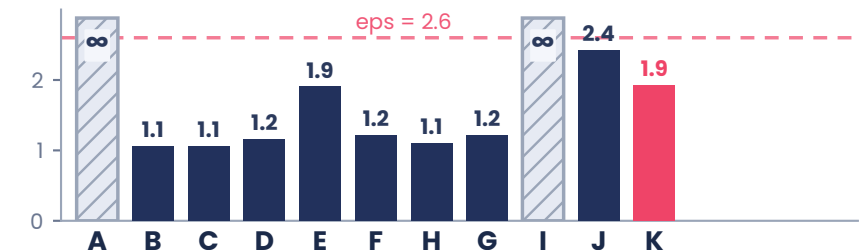
Waiting list after the update

M 1.9

L 2.0

kept (offer 2.4)
bigger, so ignored

Reachability plot (bars appear in visiting order)

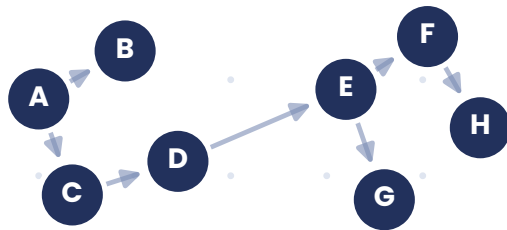


Look around point M

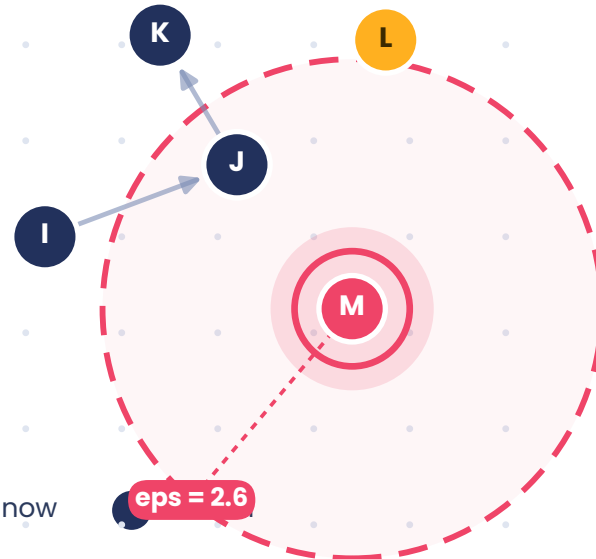
1 Visit

2 Look around

MinPts = 3 · eps = 2.6



N



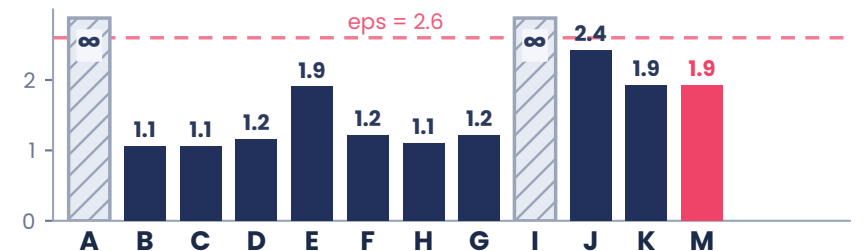
M is not a core point

Only 2 points fit inside eps, but MinPts is 3. So M cannot reach anyone else. It was reached by others, so it sits at the edge of its cluster.

Waiting list after the update

L 2.0

Reachability plot (bars appear in visiting order)

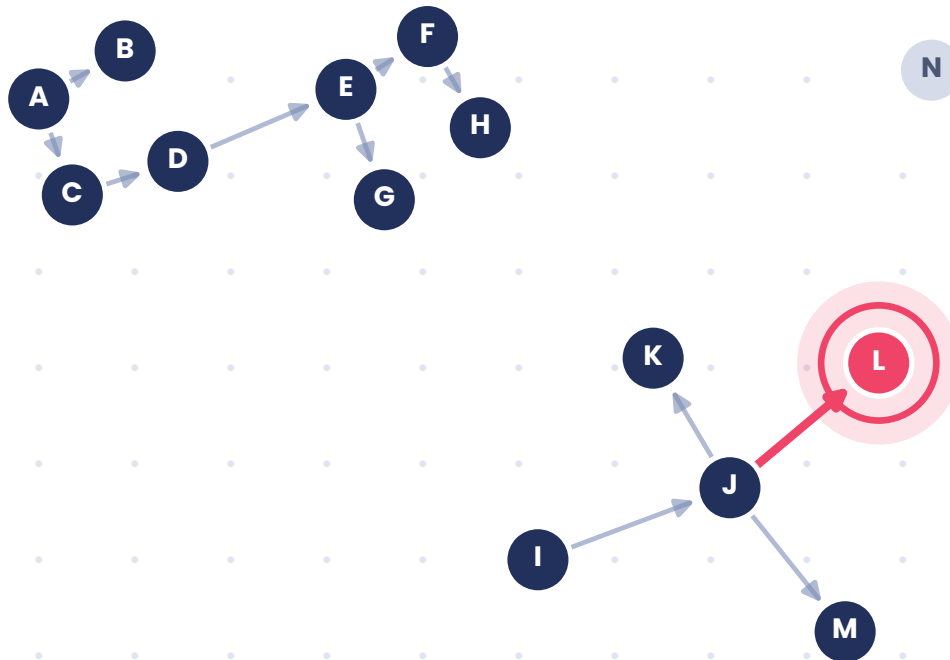


Visit point L

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



not visited waiting list visiting now visited

Visit L

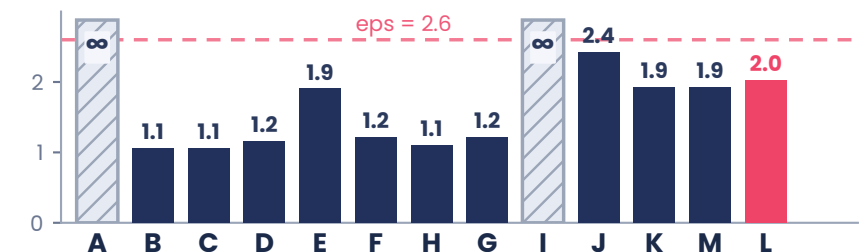
L has the smallest number in the waiting list (2.0), so we visit it next. We add a bar of height 2.0 to the plot.

Waiting list (smallest goes first)

L 2.0

next!

Reachability plot (bars appear in visiting order)

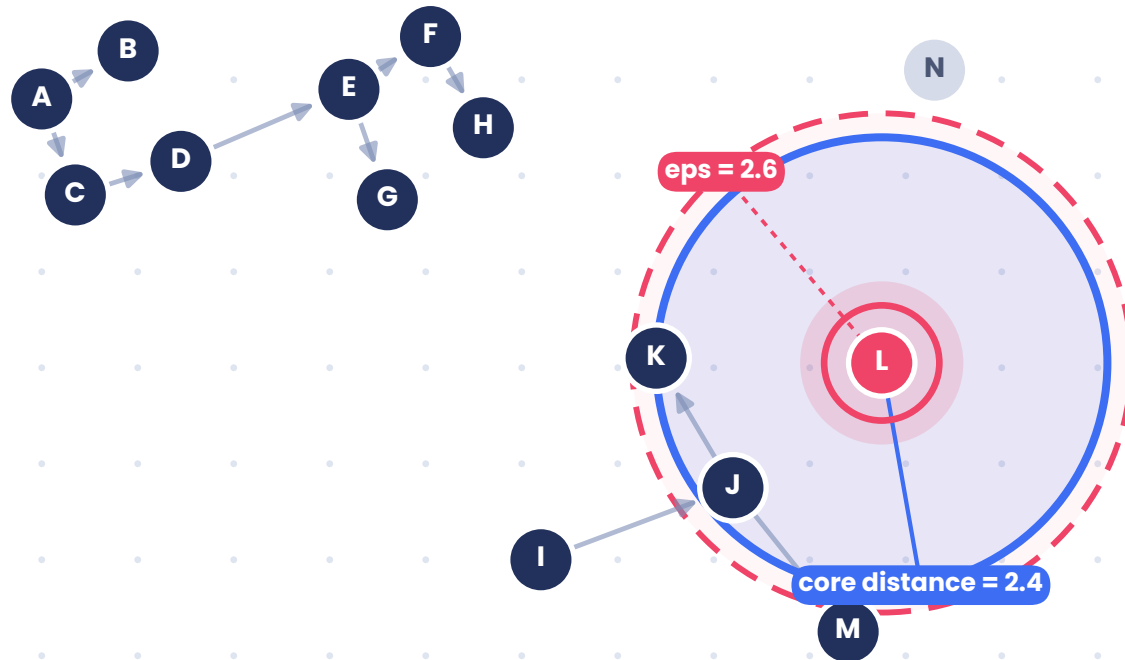


Look around point L

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited
 ● waiting list
 ● visiting now
 ● visited

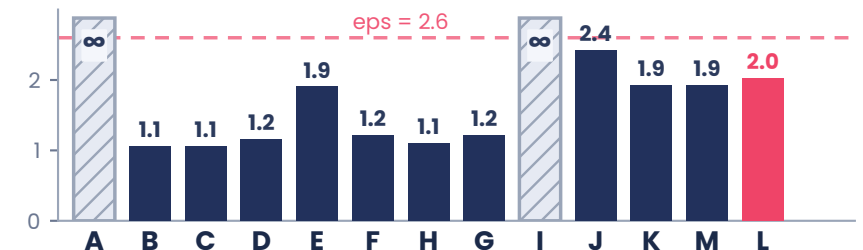
L is a core point

3 points fit inside eps (counting L). That is at least $\text{MinPts} = 3$. Core distance = 2.4. Every neighbor is already visited, so there is nothing to update.

Waiting list after the update

Empty. Nothing is waiting.

Reachability plot (bars appear in visiting order)

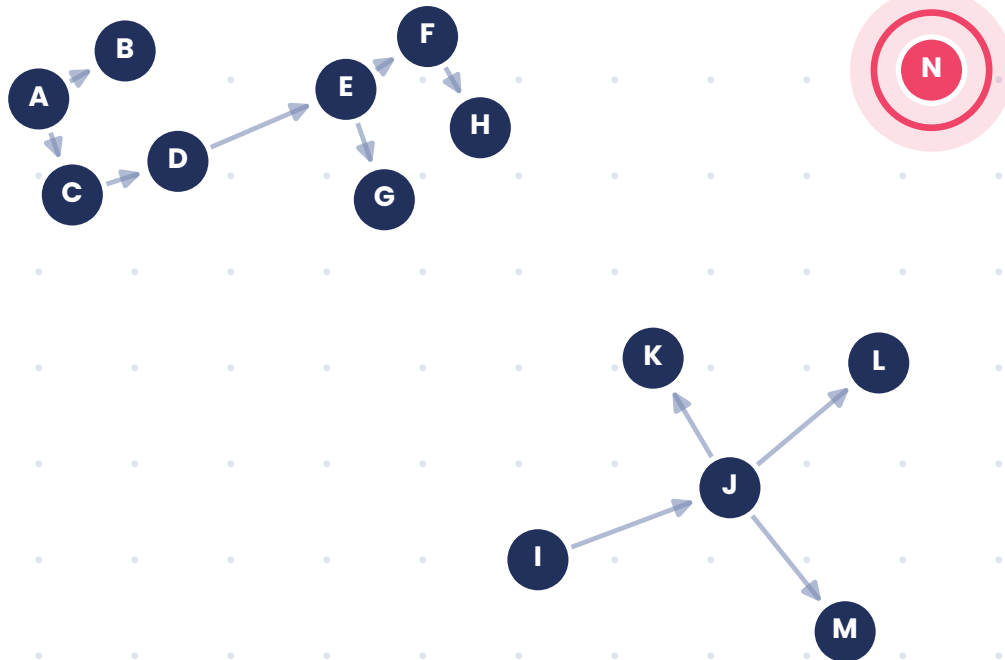


Visit point N

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



● not visited ● waiting list ● visiting now ● visited

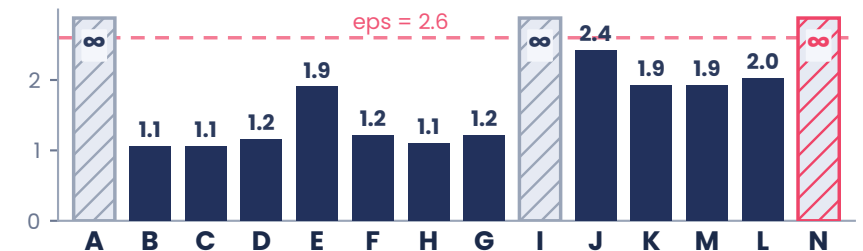
Visit N

Every other point is done. We start fresh at N. Its reachability is undefined (∞), so its bar is full height.

Waiting list (smallest goes first)

Empty. Start fresh!

Reachability plot (bars appear in visiting order)

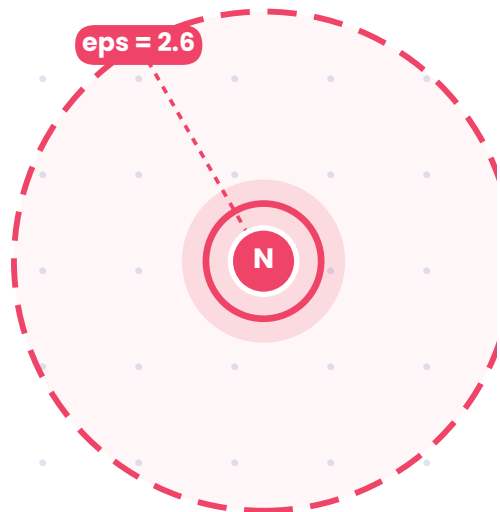
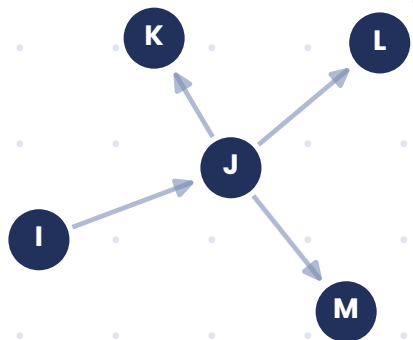
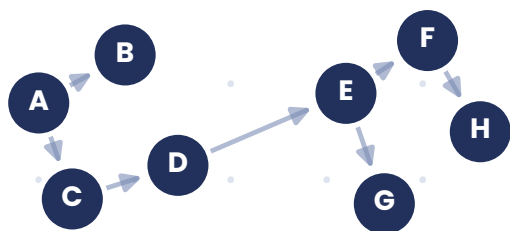


Look around point N

1 Visit

2 Look around

MinPts = 3 · eps = 2.6



not visited waiting list visiting now visited

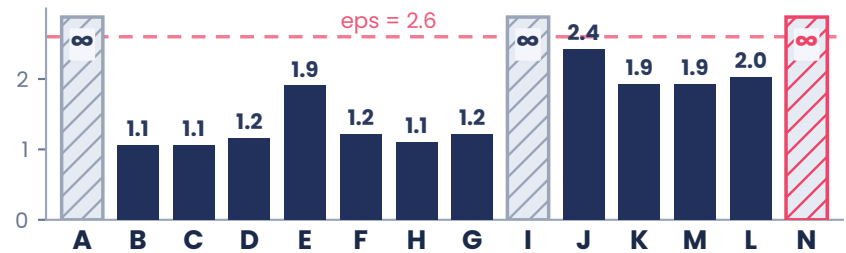
N is not a core point

Only 1 point fits inside eps, but MinPts is 3. So N cannot reach anyone else. It is far from everything: an outlier.

Waiting list after the update

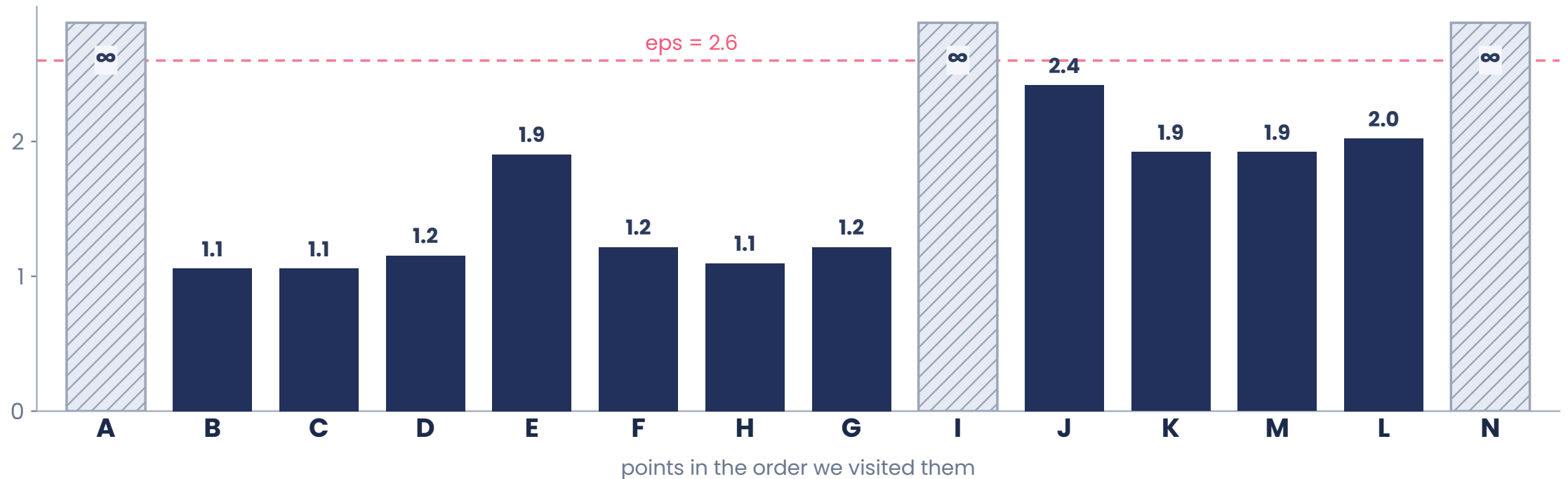
Empty. Nothing is waiting.

Reachability plot (bars appear in visiting order)



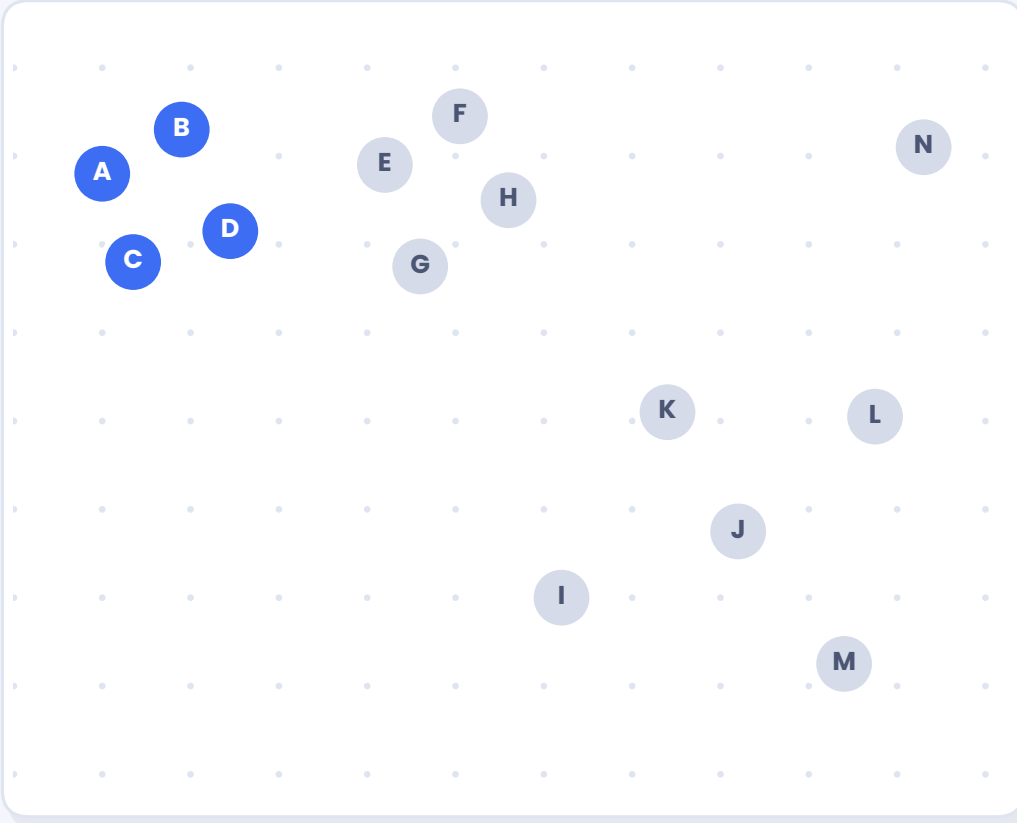
All 14 points are visited

The finished reachability plot

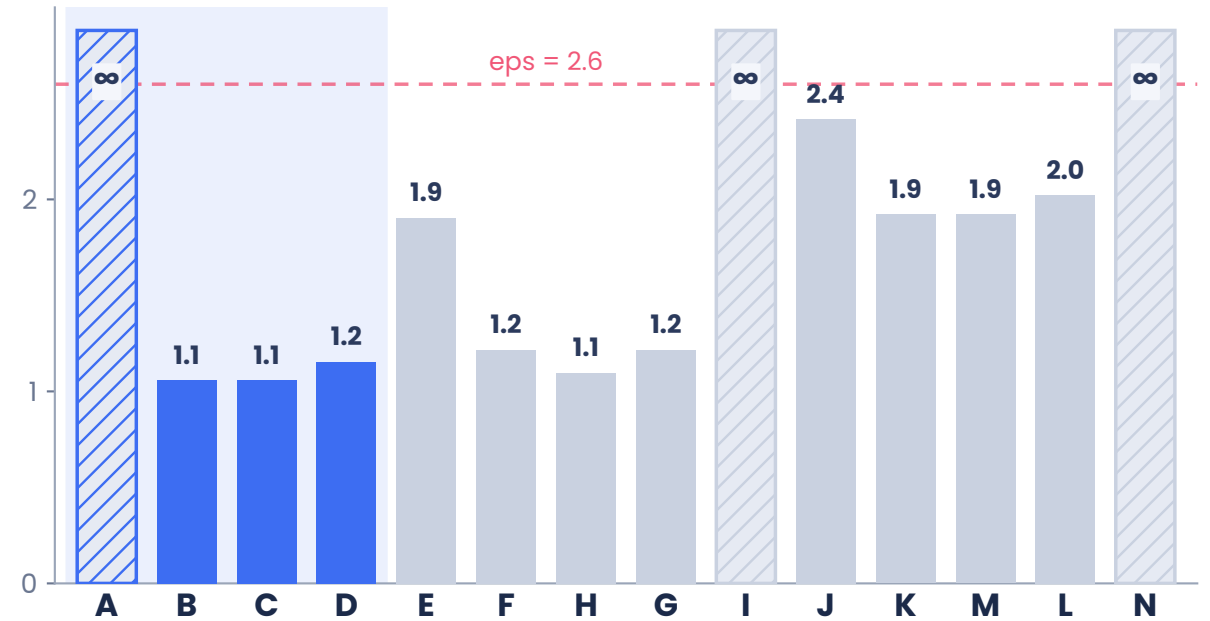


The visiting order was A, B, C, D, E, F, H, G, I, J, K, M, L, N. The bar heights are the reachability numbers we wrote down. Next, let's learn how to read this plot.

Reading the plot: low bars



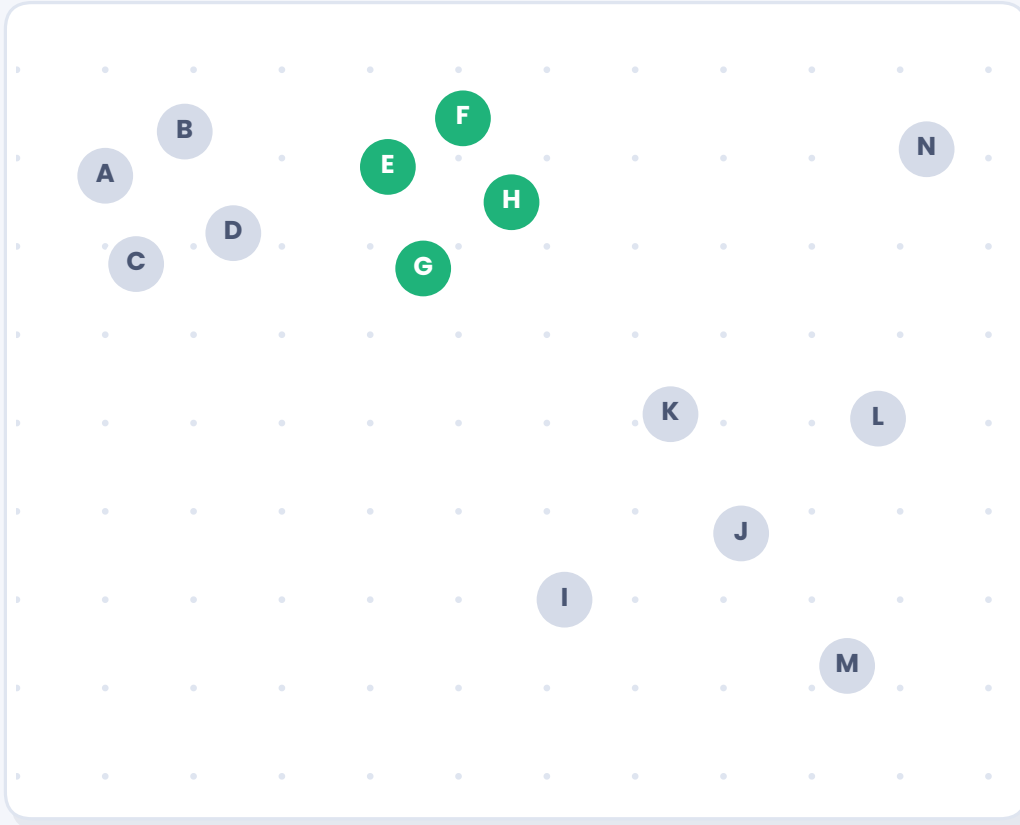
Reachability plot



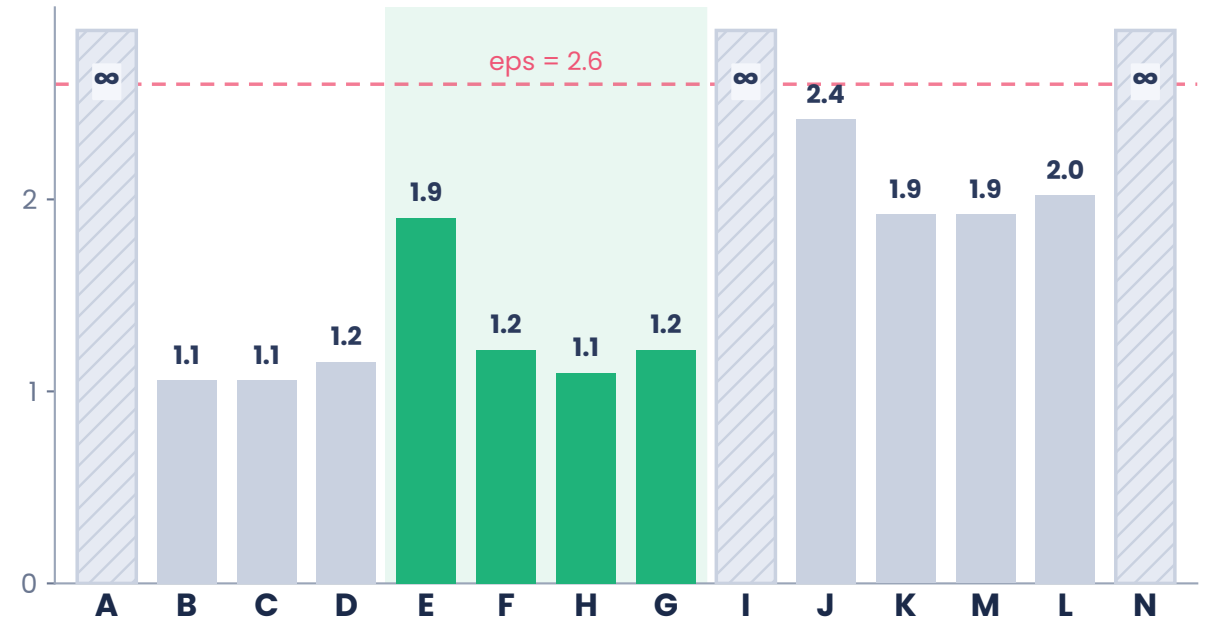
Low bars: a crowded place

The bars for A, B, C and D are low. Each point was easy to reach, so these points sit close together. A valley like this is a cluster.

Reading the plot: a peak



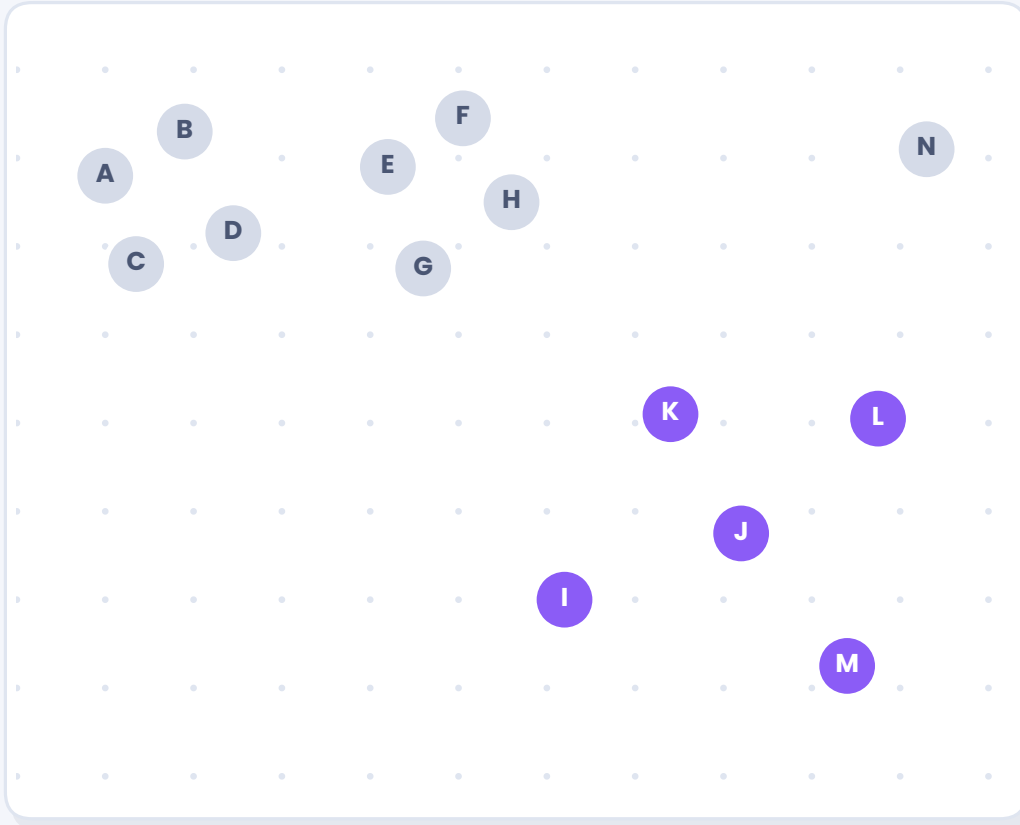
Reachability plot



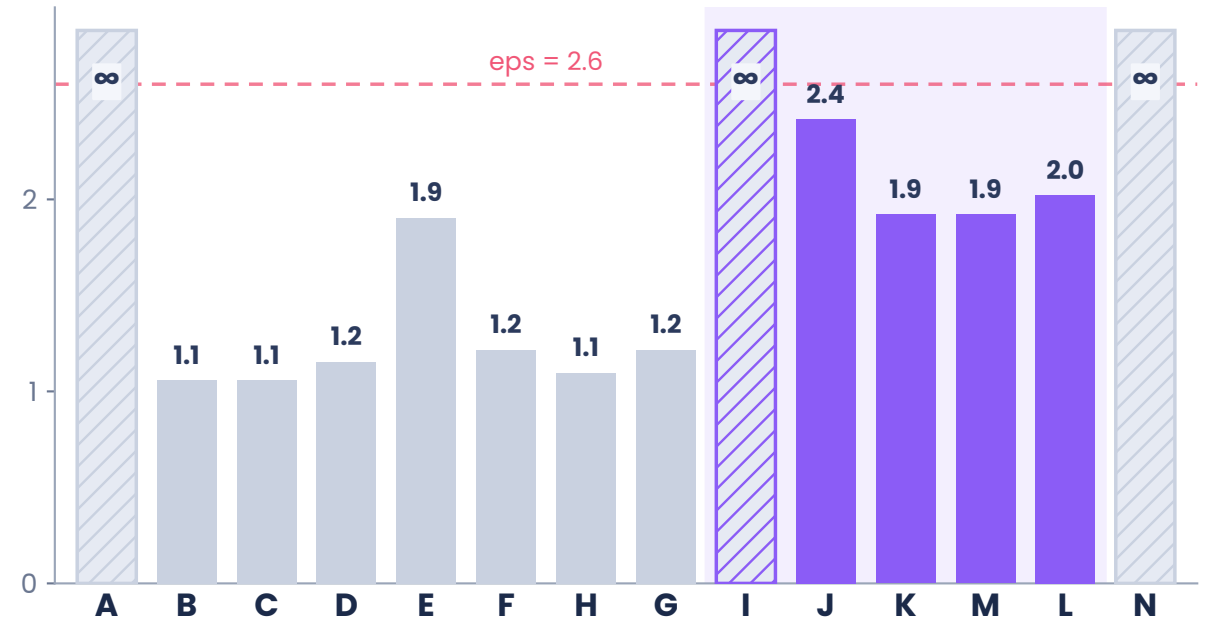
A tall bar: we crossed a gap

Bar E is tall, because we had to jump across a gap to reach it. After the peak, the bars drop again. E, F, H and G form a second cluster.

Reading the plot: a loose cluster



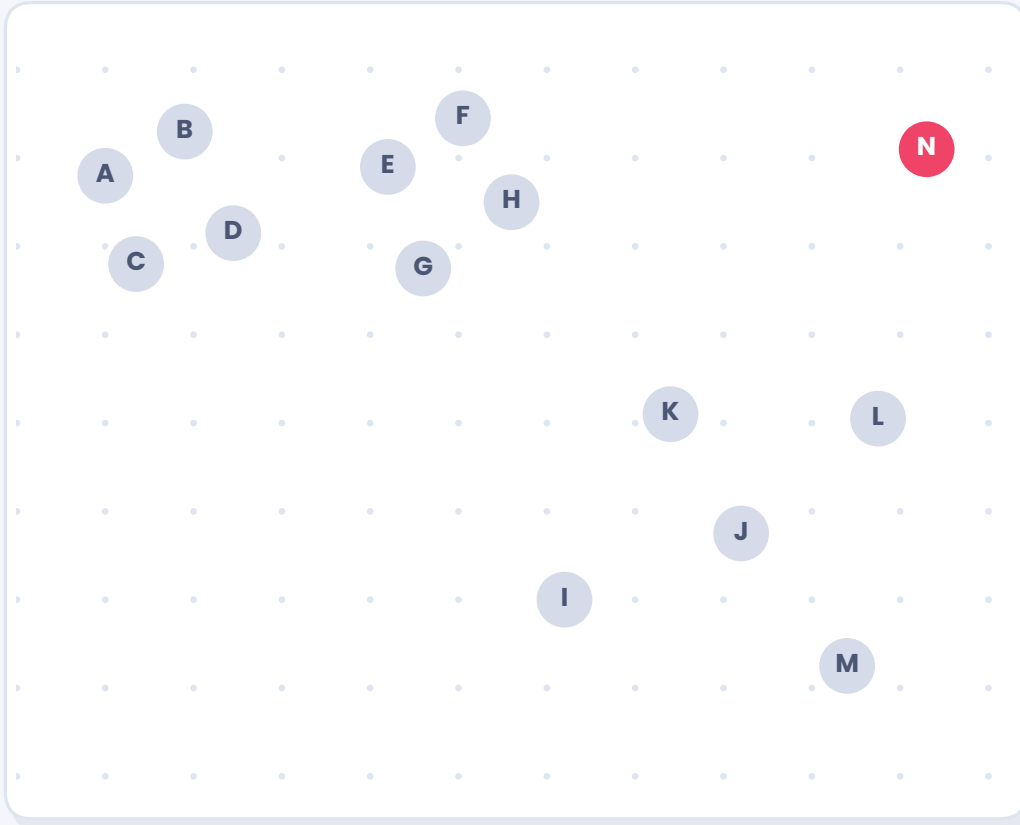
Reachability plot



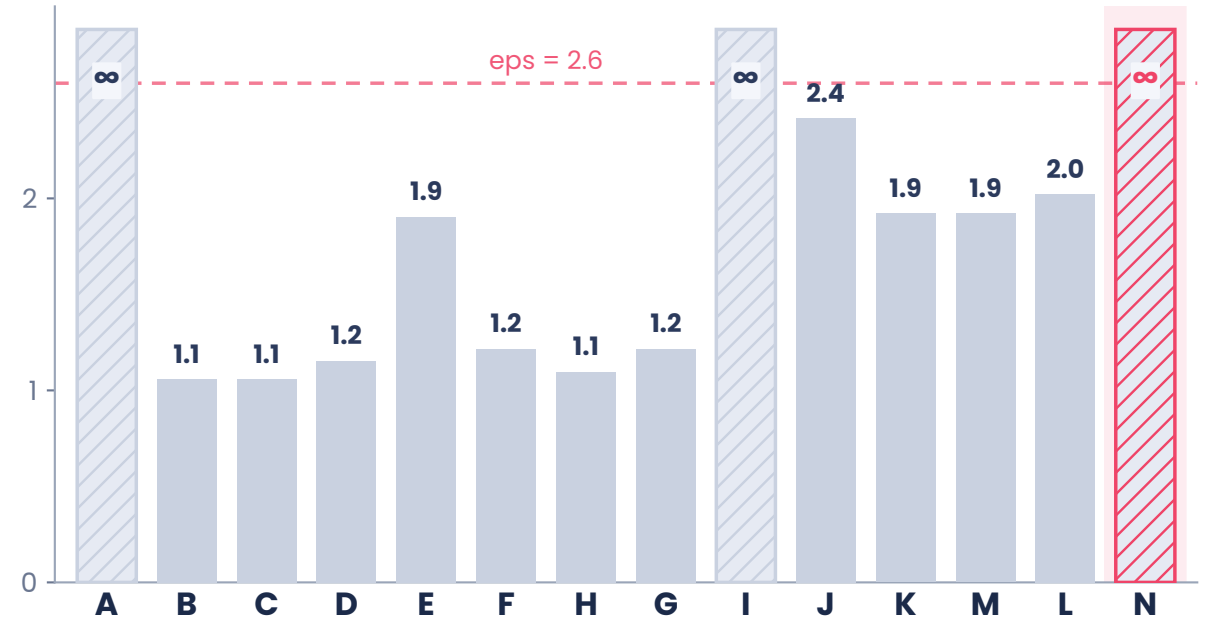
Higher bars: a loose cluster

I, J, K, M and L also form a valley, but a higher one. Their points are farther apart. Bar I is ∞ because nothing was close enough to reach it.

Reading the plot: an outlier



Reachability plot



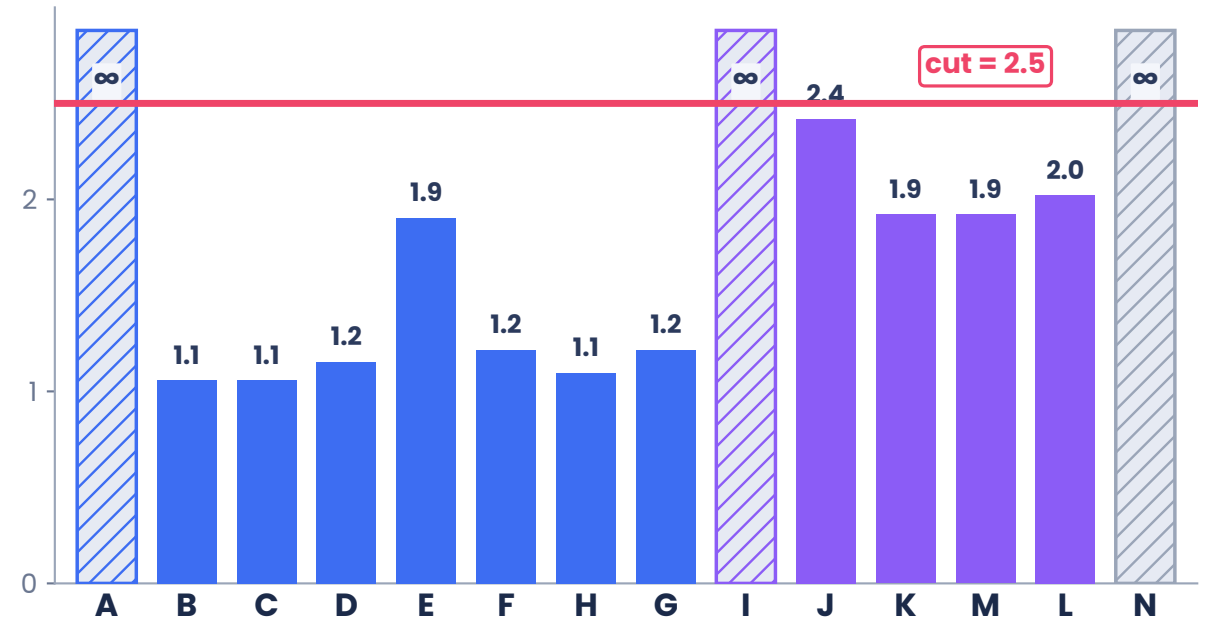
Alone: a tall bar with nothing after it

N has an ∞ bar and no valley after it. It is far from everything, so it is noise.

Cut at 2.5: a big eps



Reachability plot with a cutting line



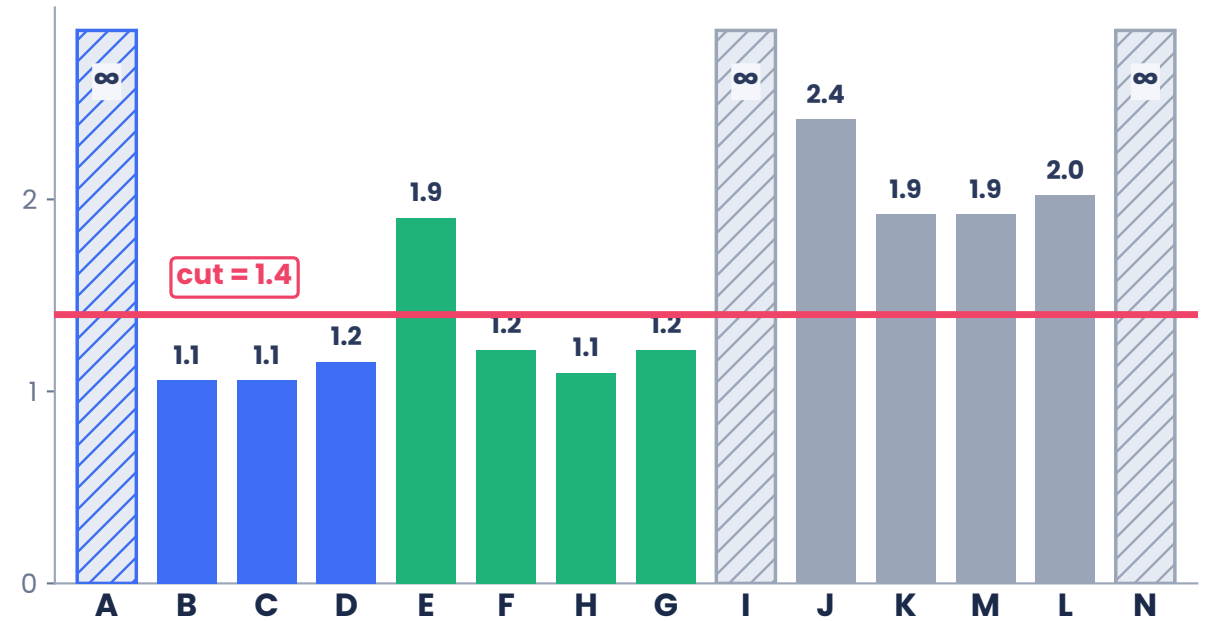
Bars below the line join a cluster

Tight groups 1 and 2 merge, and the loose group is found. This is what DBSCAN gave with a big eps.

Cut at 1.4: a small eps



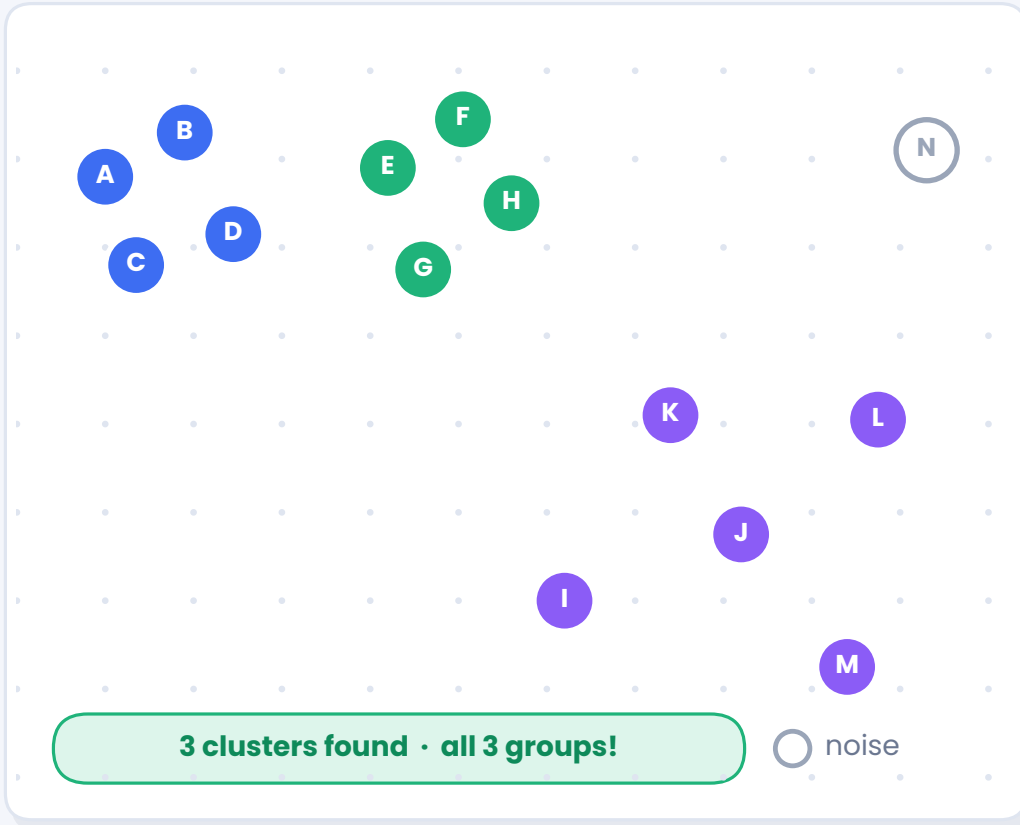
Reachability plot with a cutting line



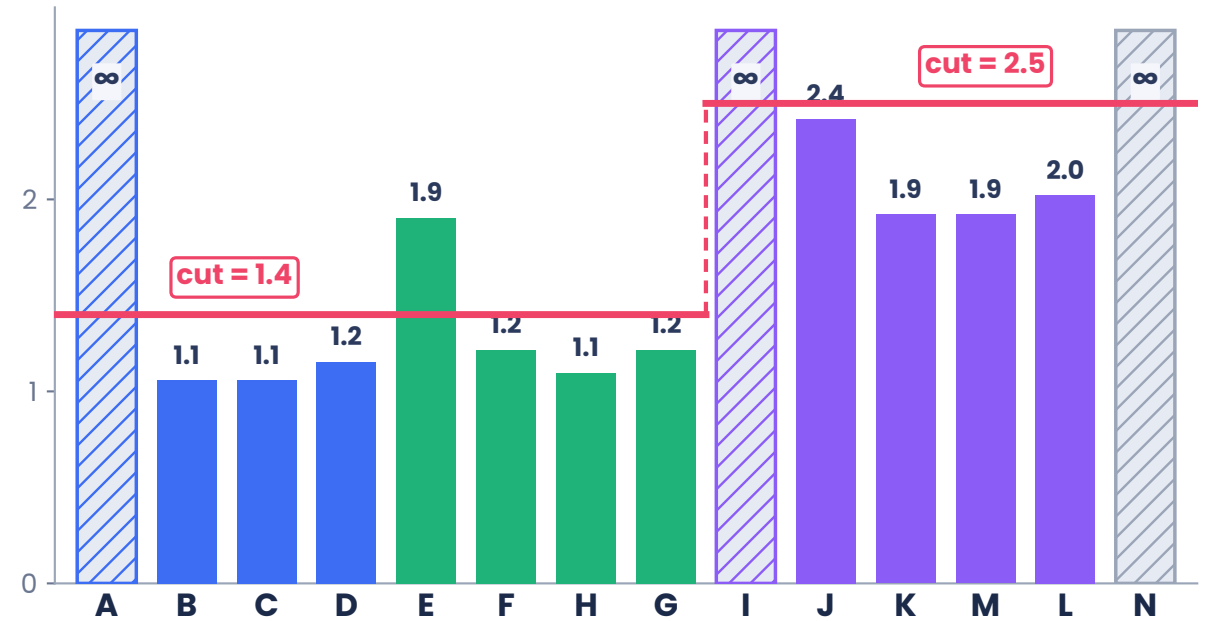
The peak at E now splits the tight groups

Bar E sticks out above the line, so the tight groups split. But the loose bars are above the line, so the loose group is lost as noise.

One cut for each valley



Reachability plot with a cutting line



Cut low in the dense area, higher in the loose area

Now all 3 groups are found. In practice, methods such as the xi method choose these cuts for us. One run of OPTICS is enough.

DBSCAN vs OPTICS

DBSCAN

OPTICS

The radius (eps)

One fixed value. It decides the result.

Only an upper limit. One run covers many radii.

Different densities

Struggles to find all groups.

Handles them well.

What we get

Cluster labels.

A visiting order and a reachability plot.

Extra work

Just the circle rule.

A waiting list and a plot to read.

Try it in Python

```
from sklearn.cluster import OPTICS

model = OPTICS(min_samples=3, max_eps=2.6)
model.fit(points)

model.ordering_      # visiting order
model.reachability_  # bar heights
model.labels_        # clusters, -1 = noise
```

What the names mean

min_samples

= MinPts

max_eps

= eps

reachability_

undefined shows as inf

On our 14 points, scikit-learn gives the same visiting order and bar heights as our hand simulation, and it finds the same 3 clusters and 1 noise point.

Key takeaways

Why OPTICS?

DBSCAN uses one eps. That fails when groups have different densities.

The method

OPTICS visits points in a smart order and writes down a reachability number for each.

The numbers

Core distance = the smallest circle with MinPts points. Reachability = $\max(\text{core distance}, \text{distance})$.

The plot

Valleys are clusters, peaks are gaps. Cut at different heights to find all groups.

Quick quiz

1

What problem with DBSCAN does OPTICS fix?

2

What is the core distance?

3

How do we compute a reachability distance?

4

What do valleys and peaks mean in the plot?

Quick quiz: answers

1

What problem with DBSCAN does OPTICS fix?

DBSCAN uses one eps, so it struggles when groups have different densities. OPTICS does not force one radius.

2

What is the core distance?

The smallest radius of a circle around a point that holds MinPts points, counting the point itself.

3

How do we compute a reachability distance?

Take the larger of two numbers: the core distance of the current point, and the distance to the neighbor.

4

What do valleys and peaks mean in the plot?

Valleys are clusters, because the points are close together. Peaks are gaps between clusters.